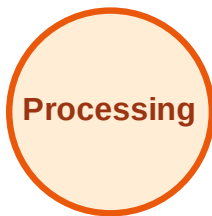


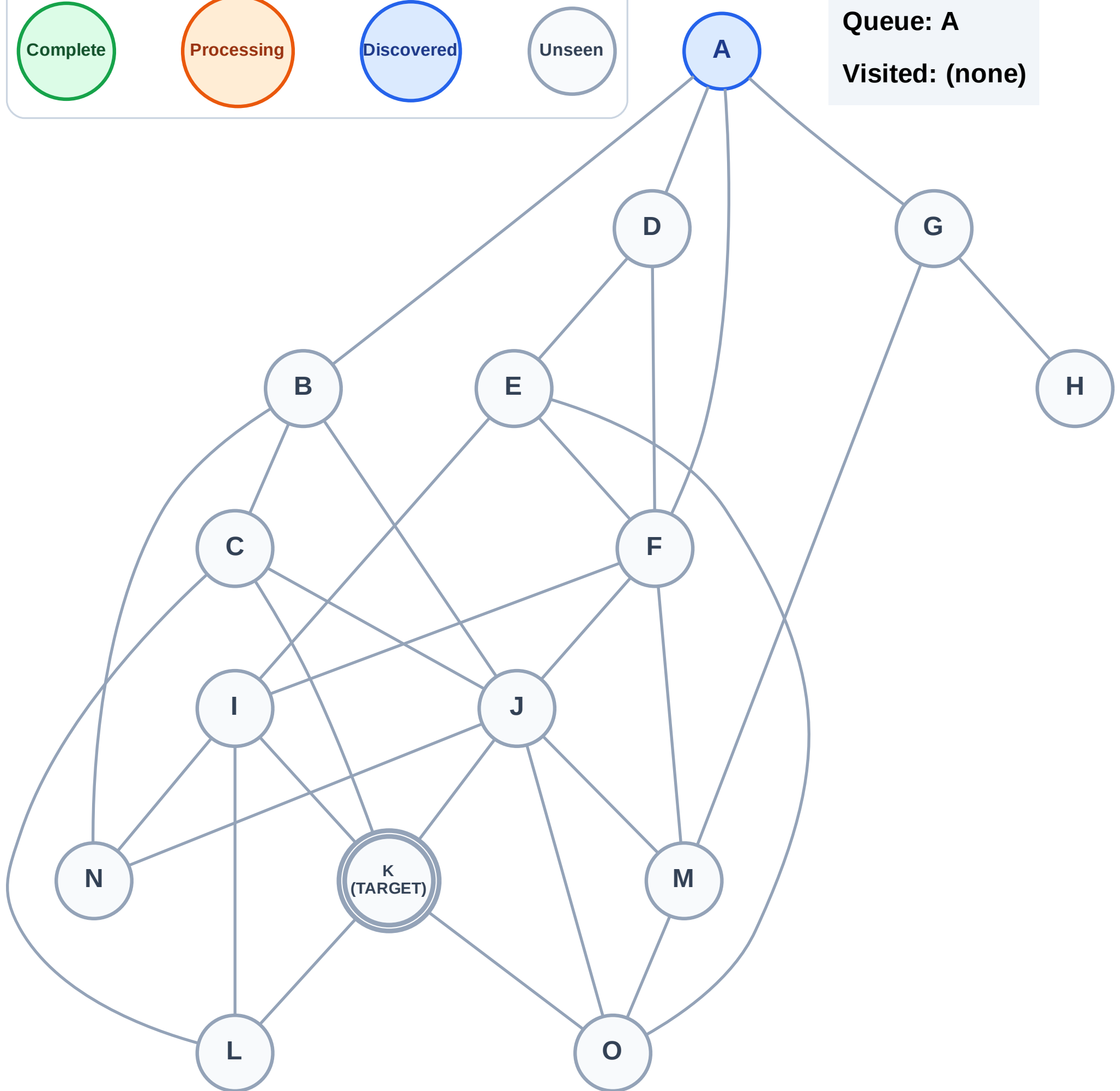
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



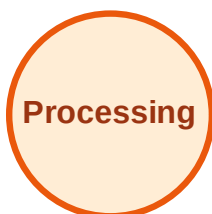
Queue: A
Visited: (none)



BFS Traversal

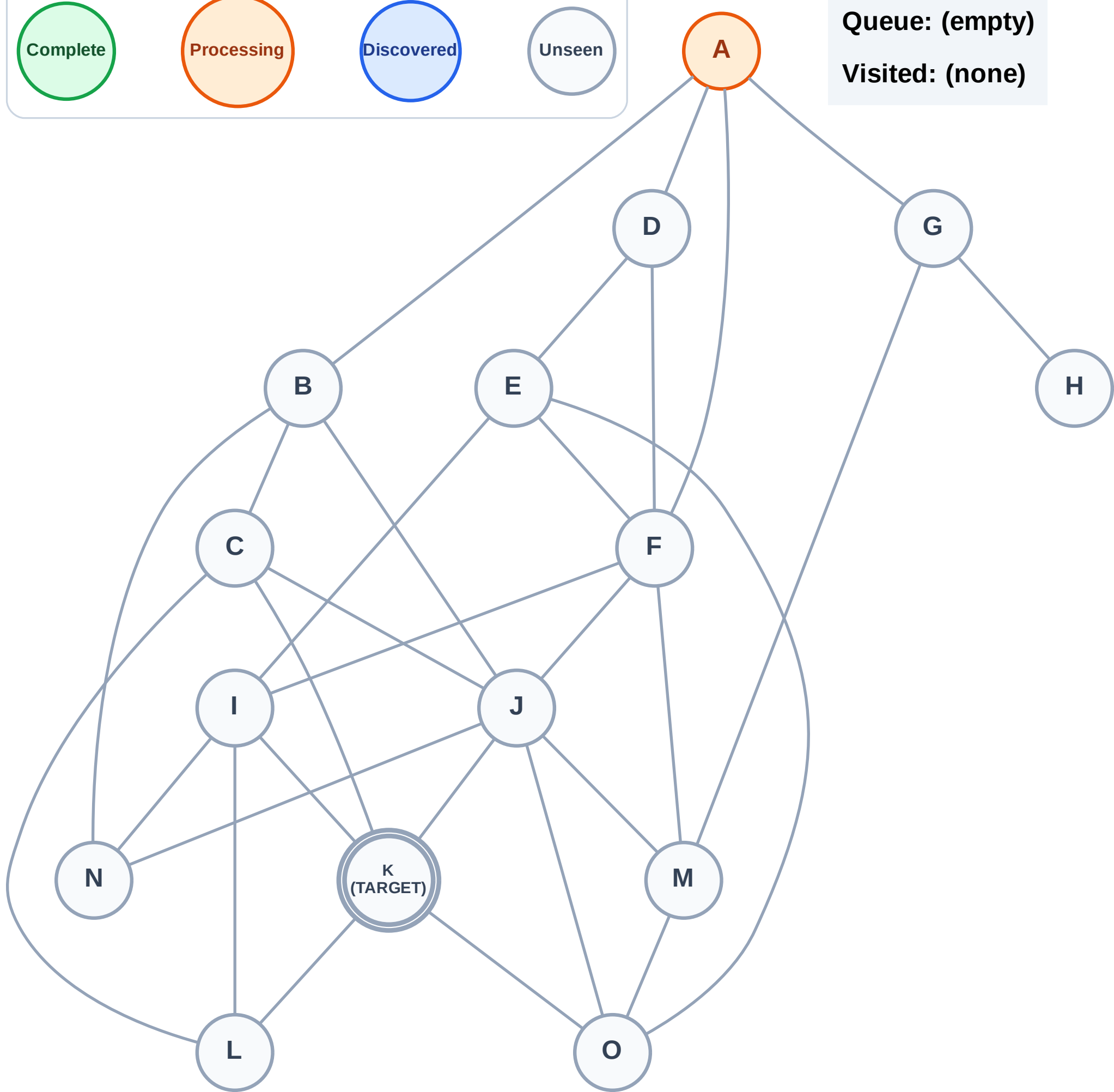
Step 2: Process node A

Legend



Queue: (empty)

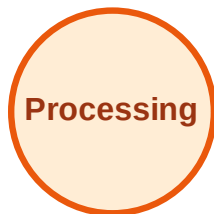
Visited: (none)



BFS Traversal

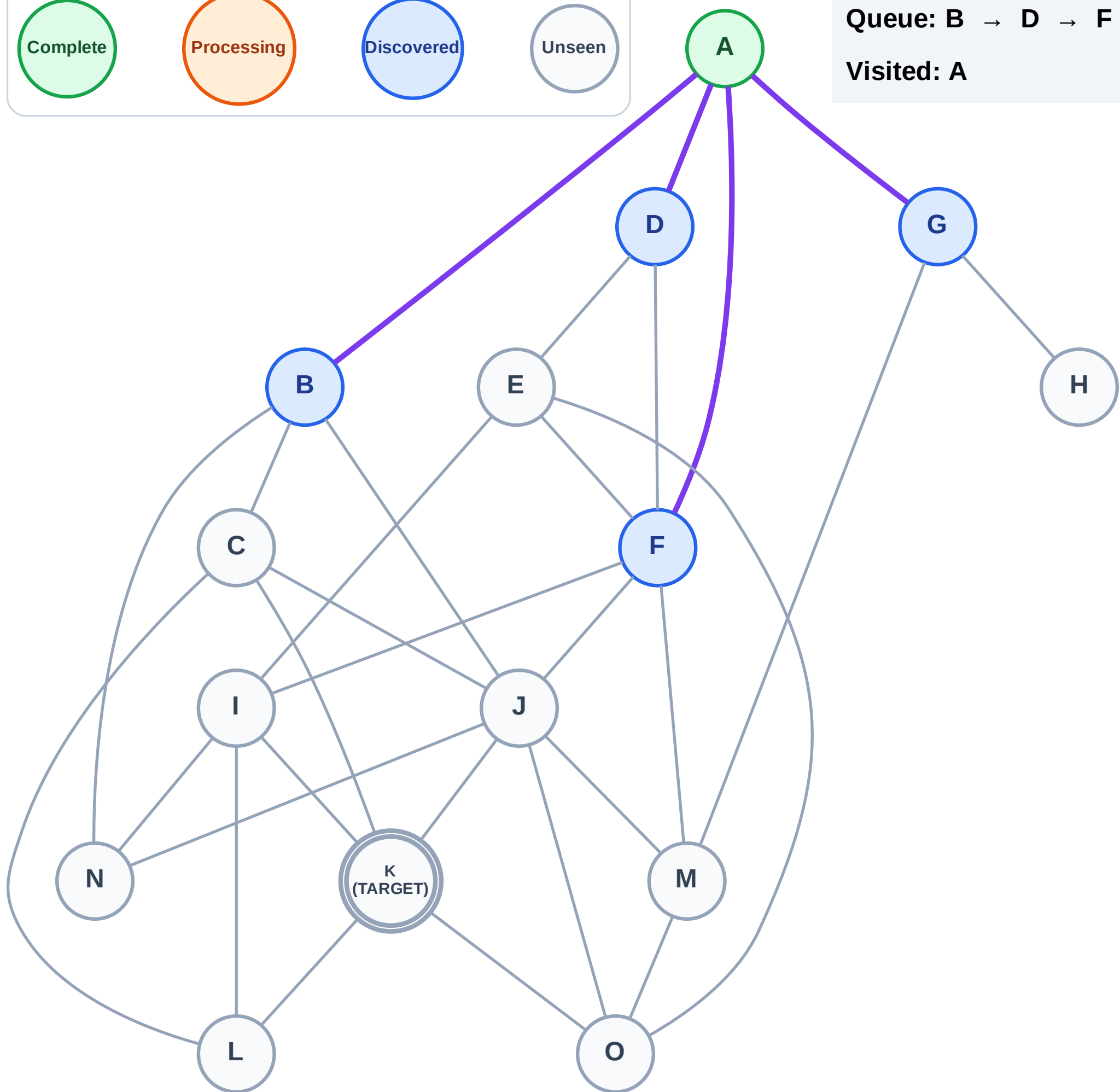
Step 3: Discover B, D, F, G from A

Legend



Queue: B → D → F → G

Visited: A



BFS Traversal

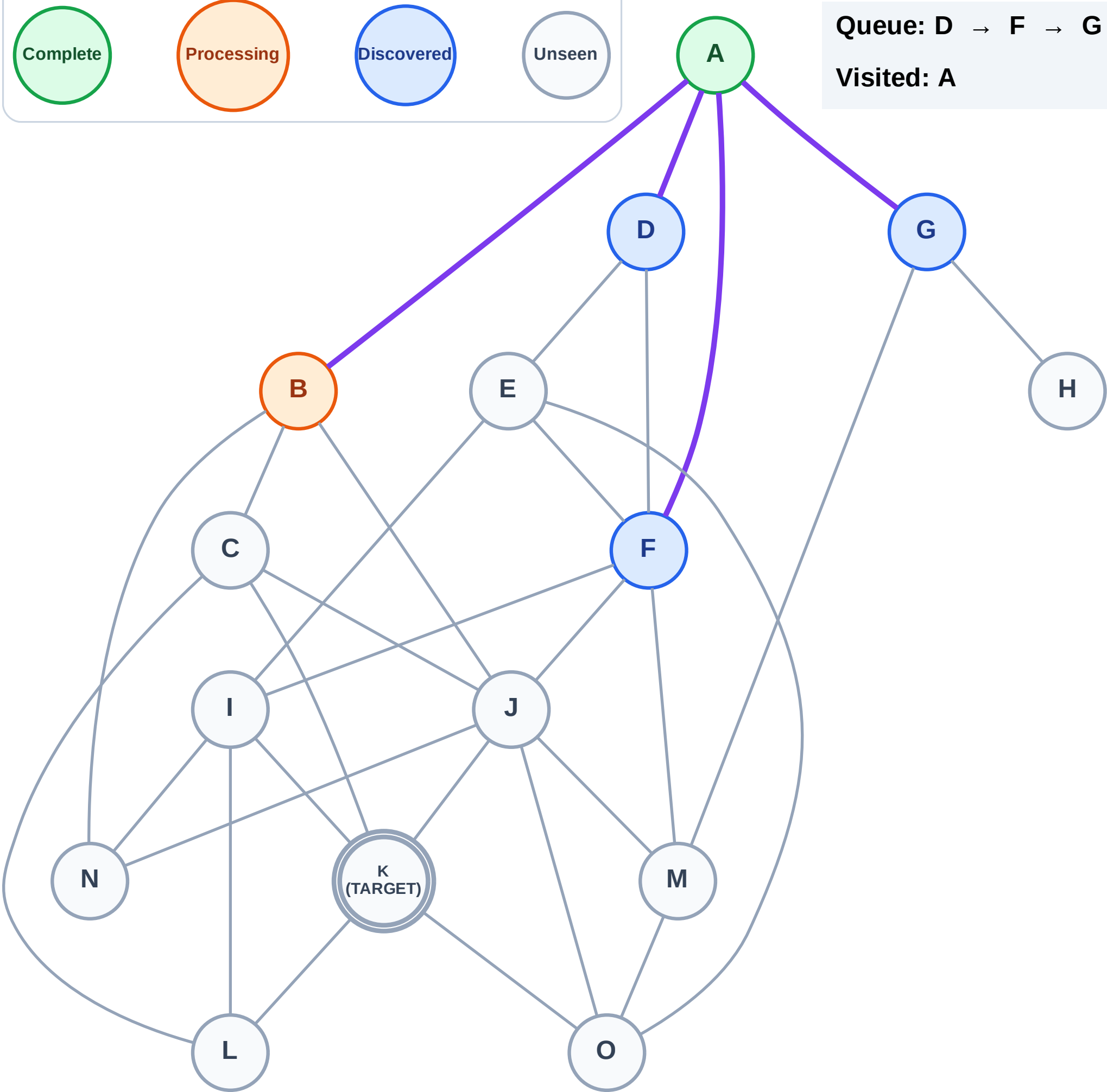
Step 4: Process node B

Legend



Queue: D → F → G

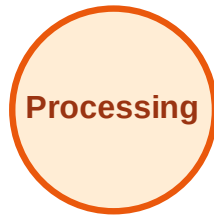
Visited: A



BFS Traversal

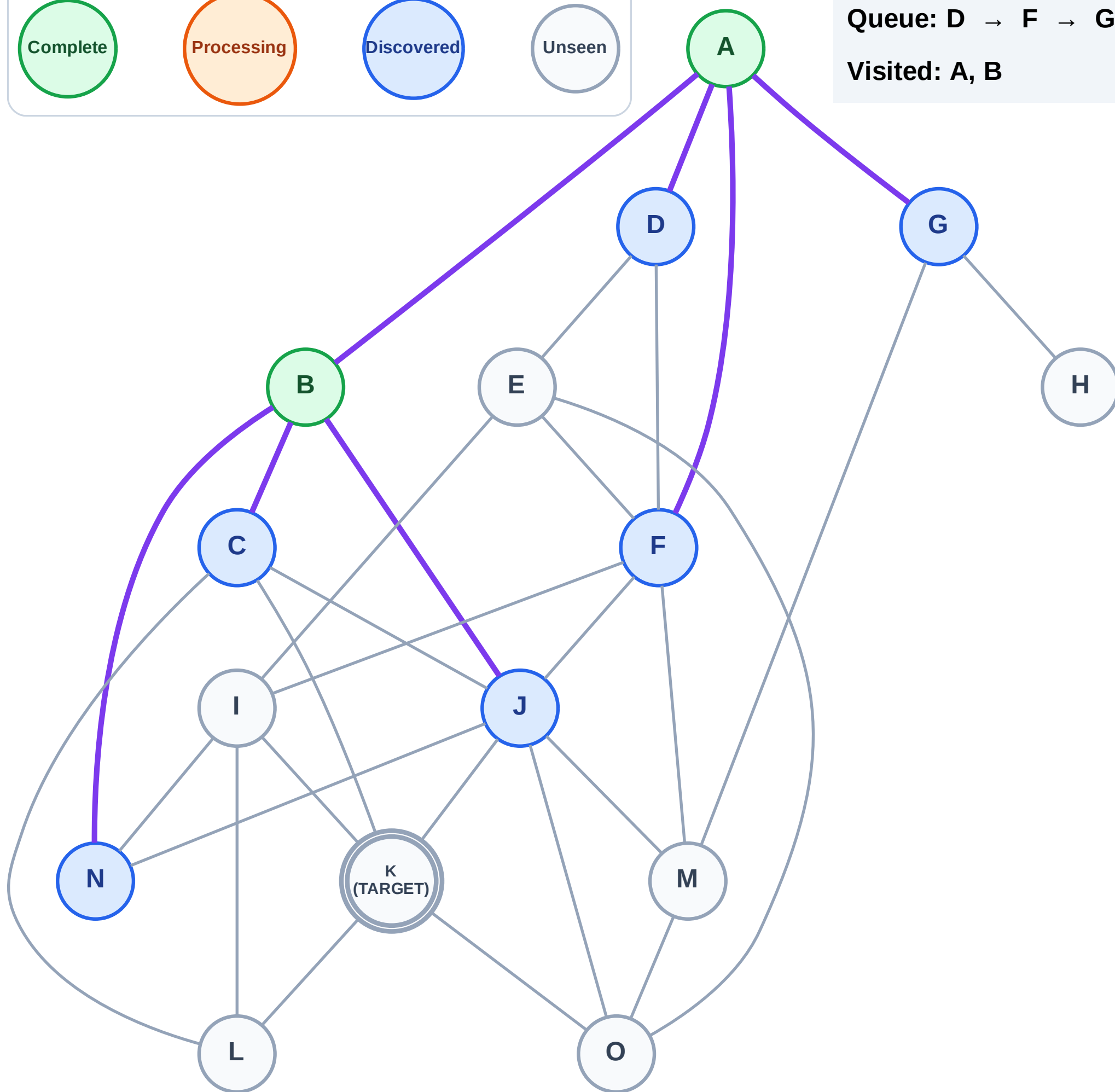
Step 5: Discover C, J, N from B

Legend



Queue: D → F → G → C → J → N

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

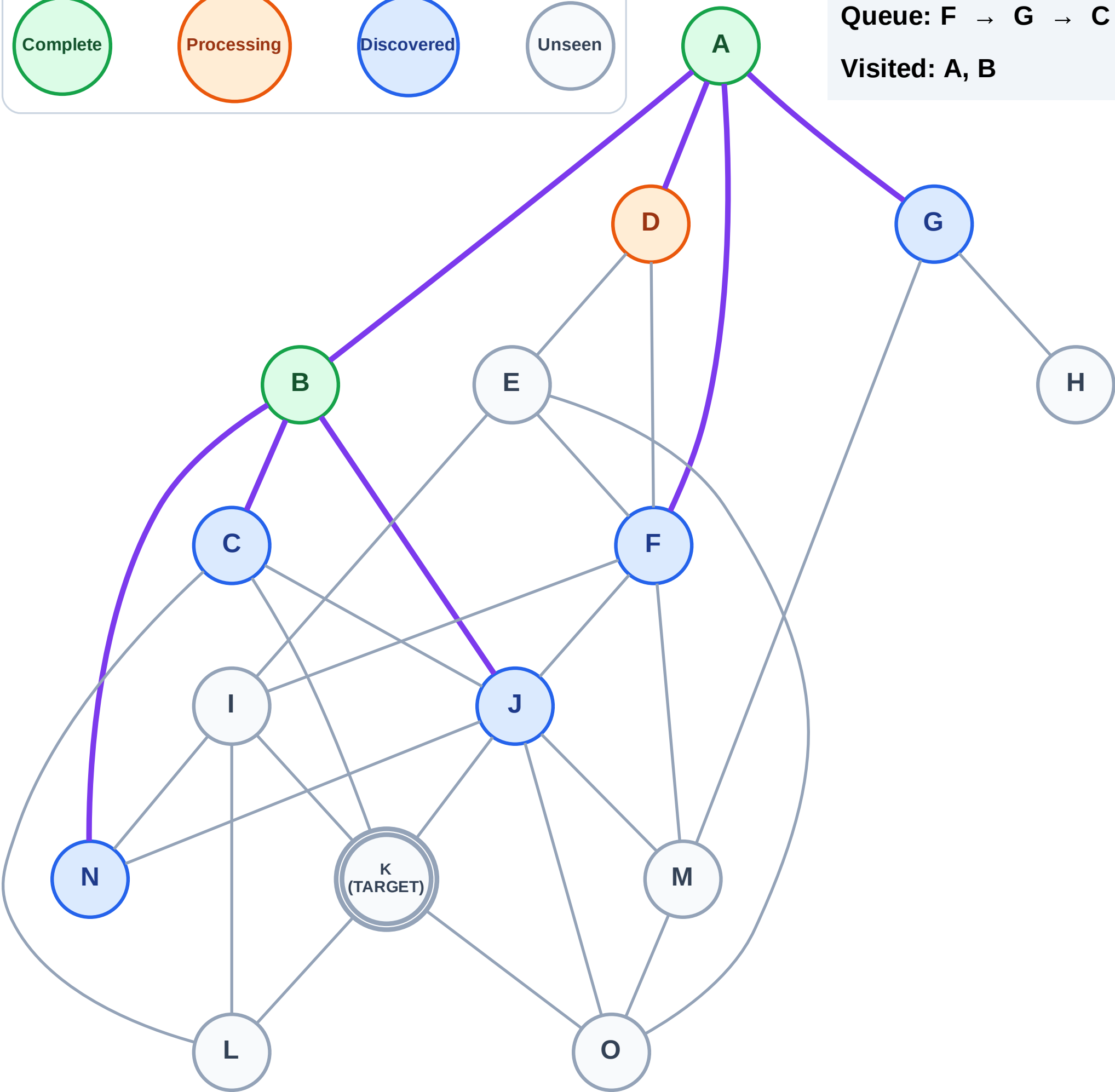
Processing

Discovered

Unseen

Queue: F → G → C → J → N

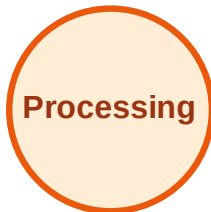
Visited: A, B



BFS Traversal

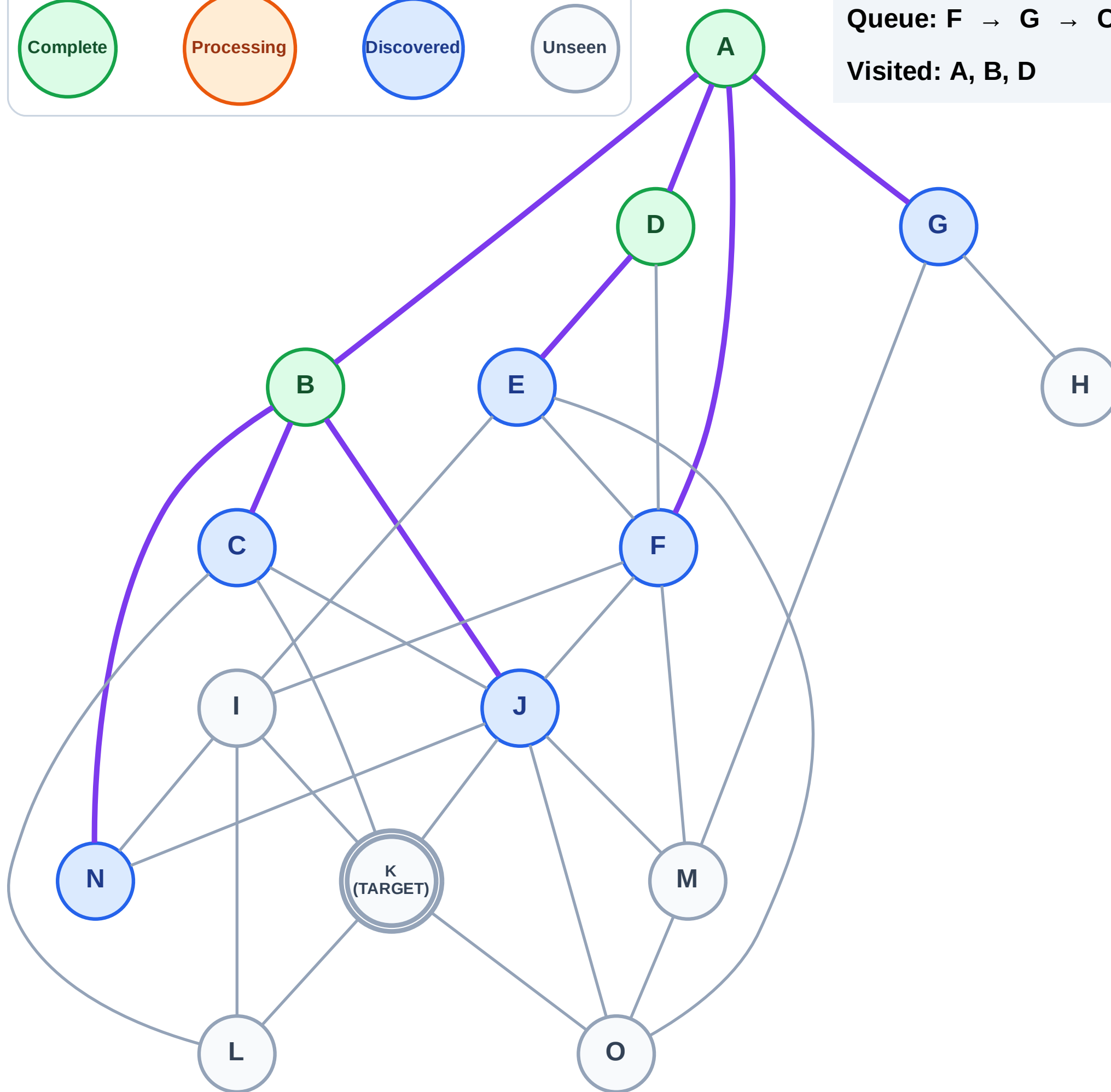
Step 7: Discover E from D

Legend



Queue: F → G → C → J → N → E

Visited: A, B, D



BFS Traversal

Step 8: Process node F

Legend

Complete

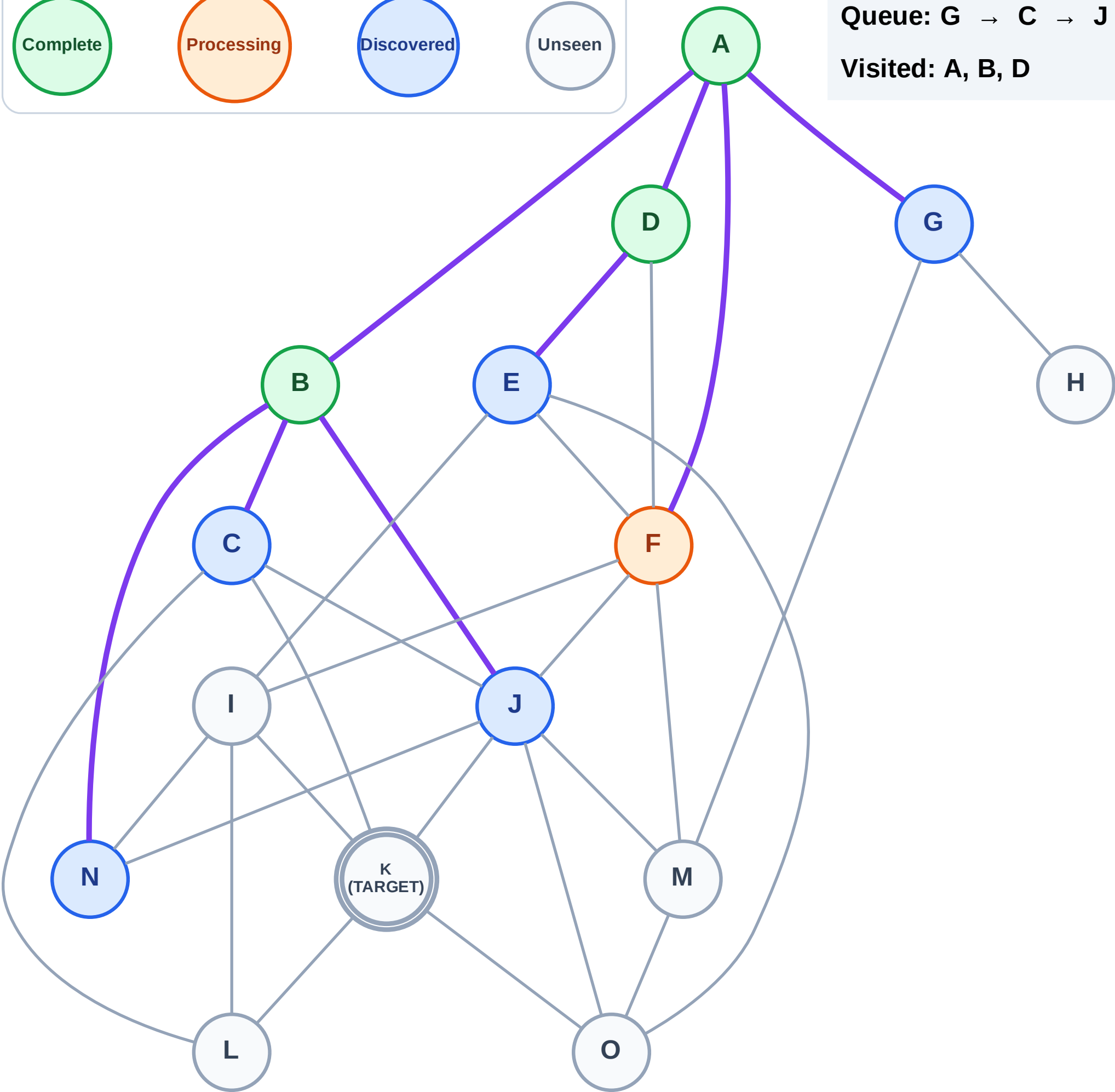
Processing

Discovered

Unseen

Queue: G → C → J → N → E

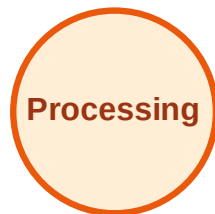
Visited: A, B, D



BFS Traversal

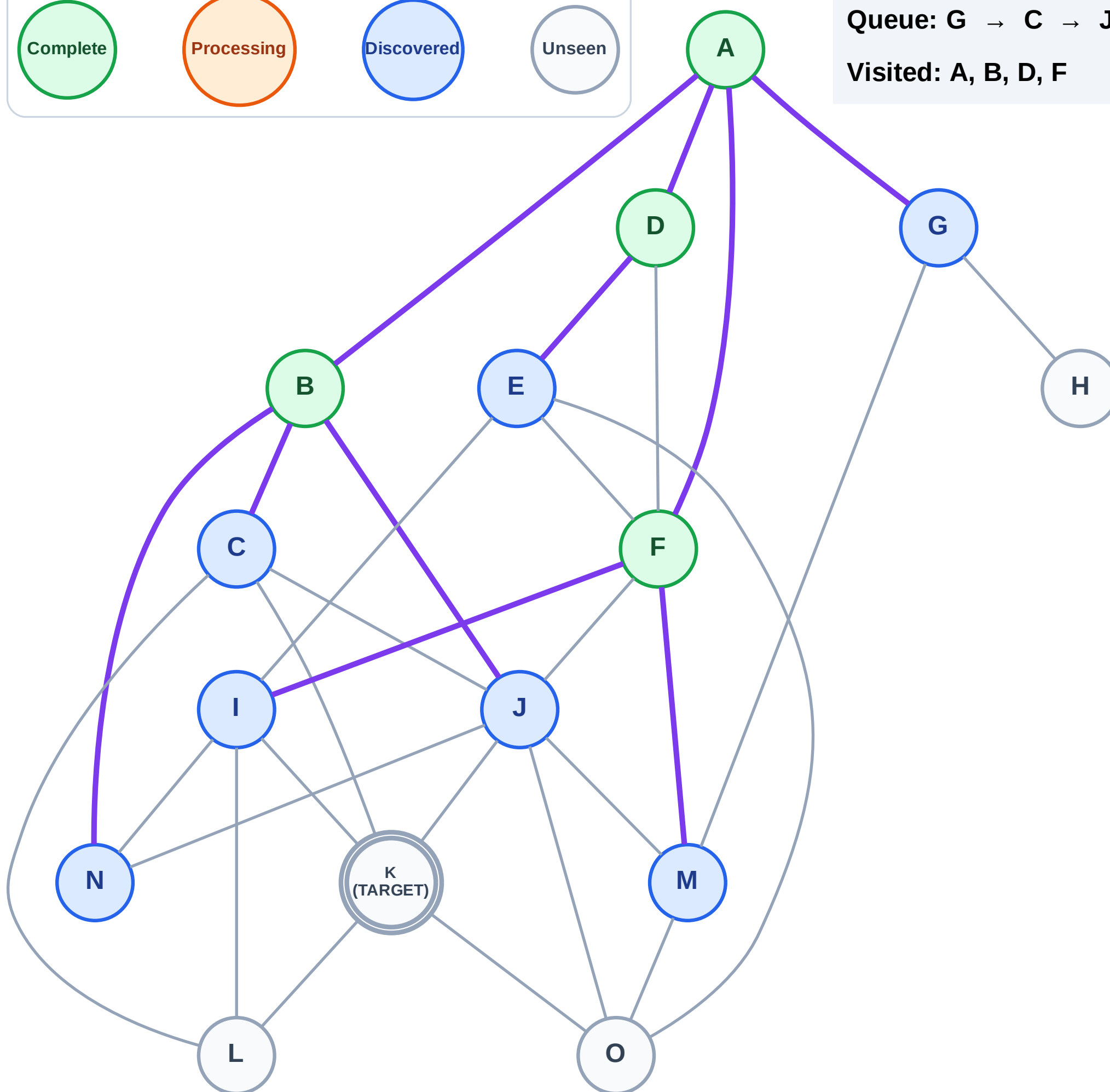
Step 9: Discover I, M from F

Legend



Queue: G → C → J → N → E → I → M

Visited: A, B, D, F



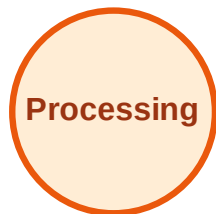
BFS Traversal

Step 10: Process node G

Legend



Complete



Processing



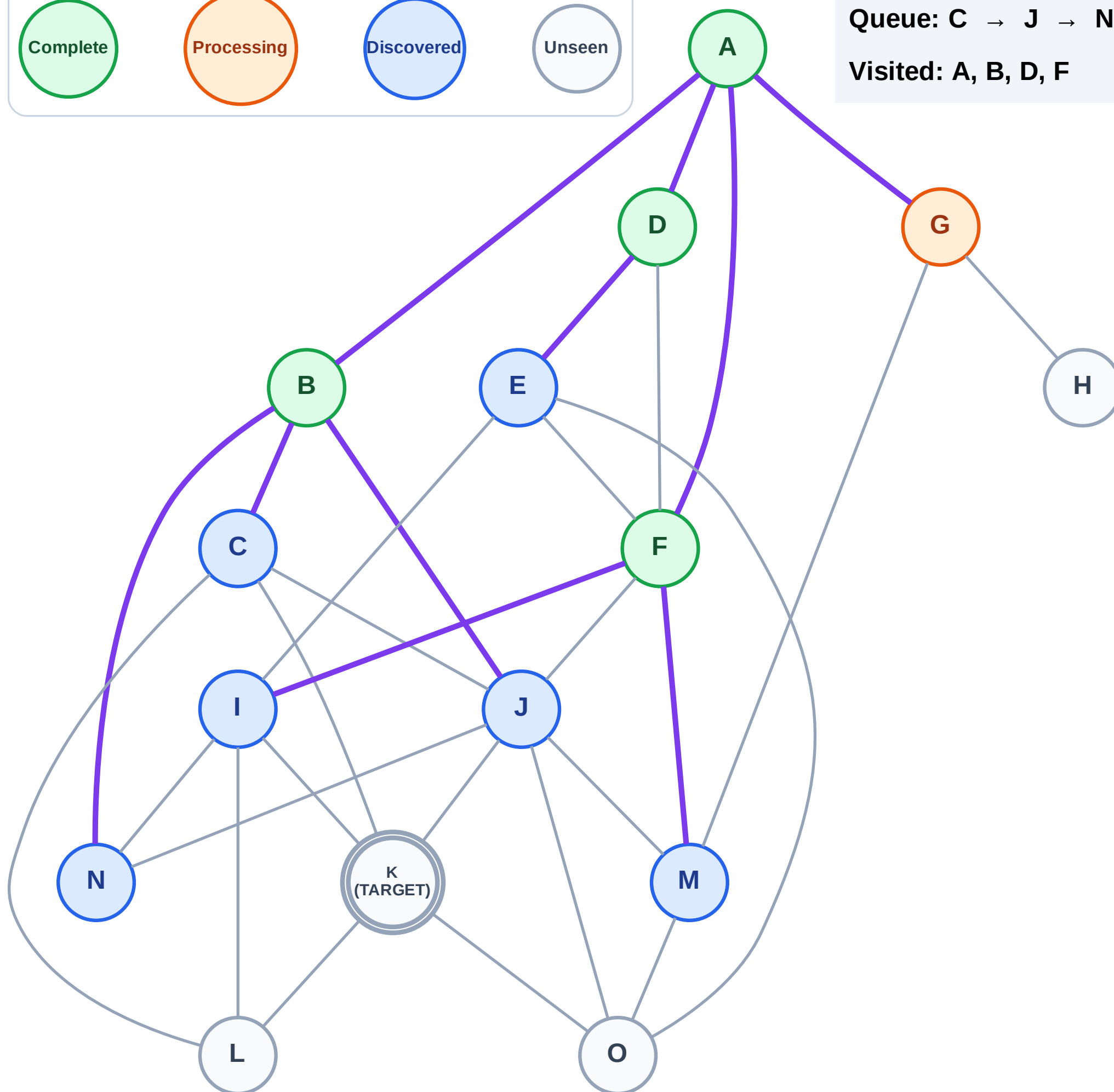
Discovered



Unseen

Queue: C → J → N → E → I → M

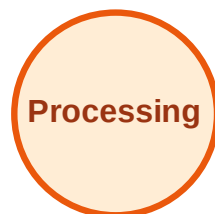
Visited: A, B, D, F



BFS Traversal

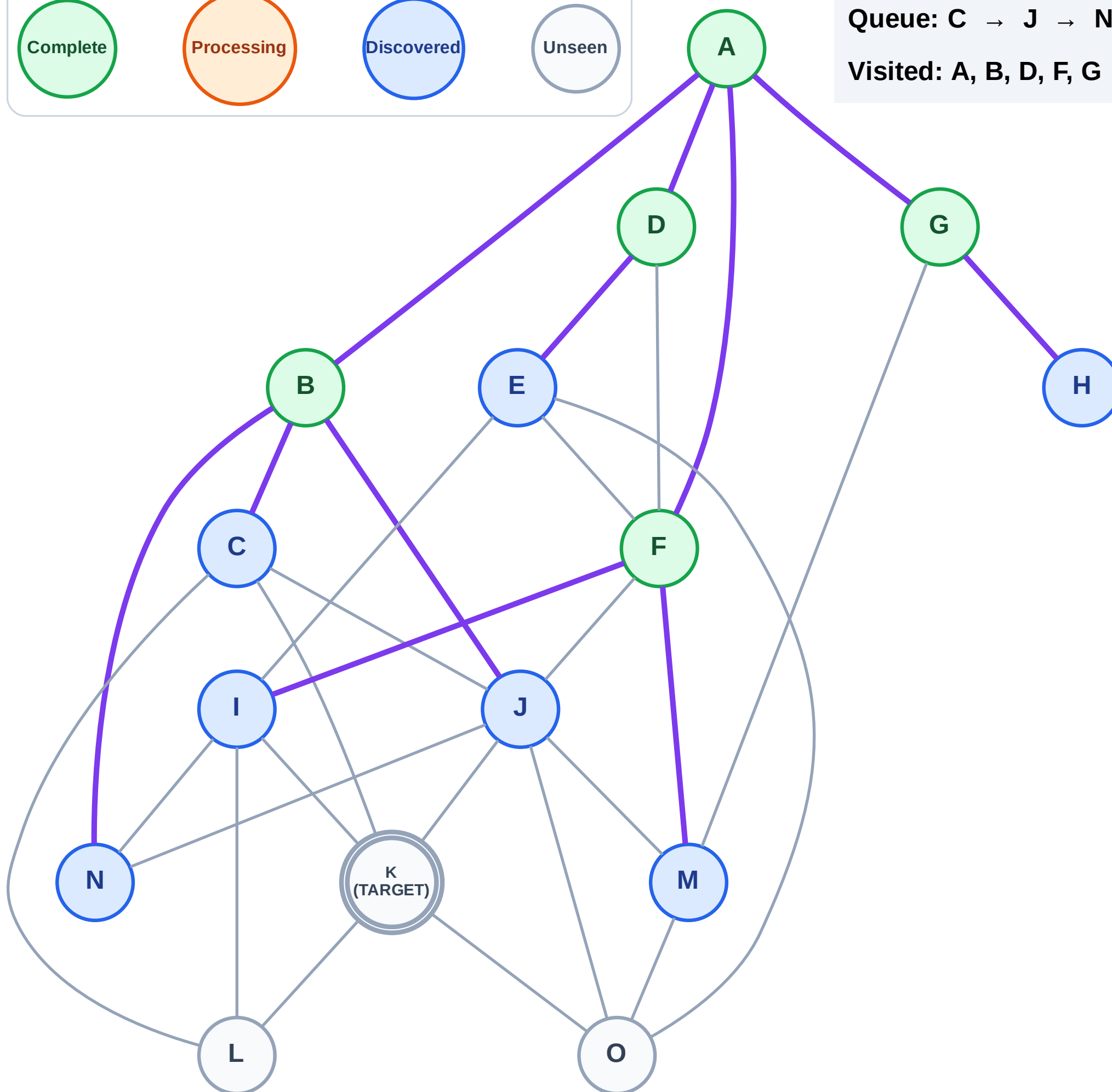
Step 11: Discover H from G

Legend



Queue: C → J → N → E → I → M → H

Visited: A, B, D, F, G



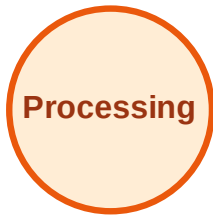
BFS Traversal

Step 12: Process node C

Legend



Complete



Processing



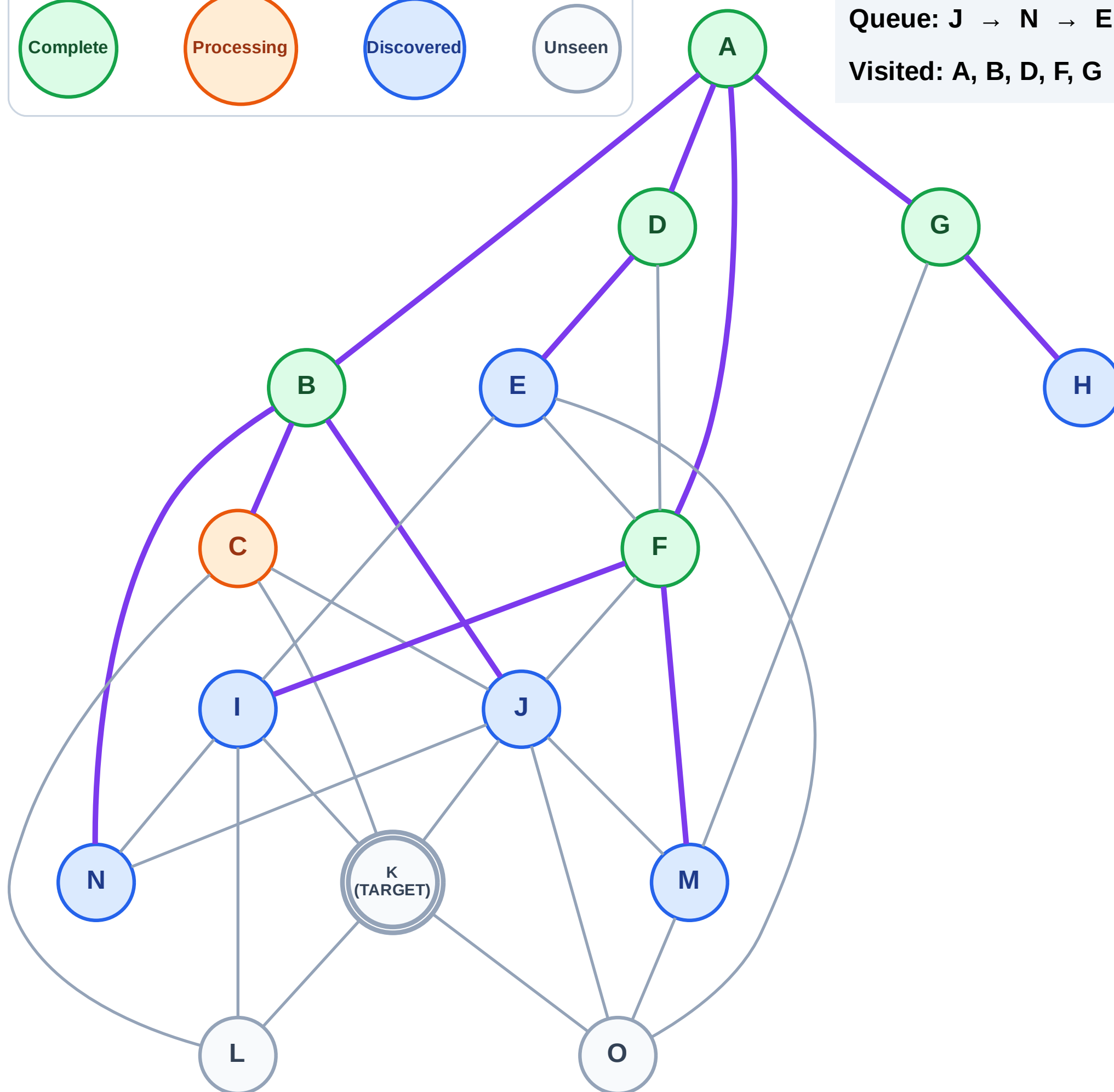
Discovered



Unseen

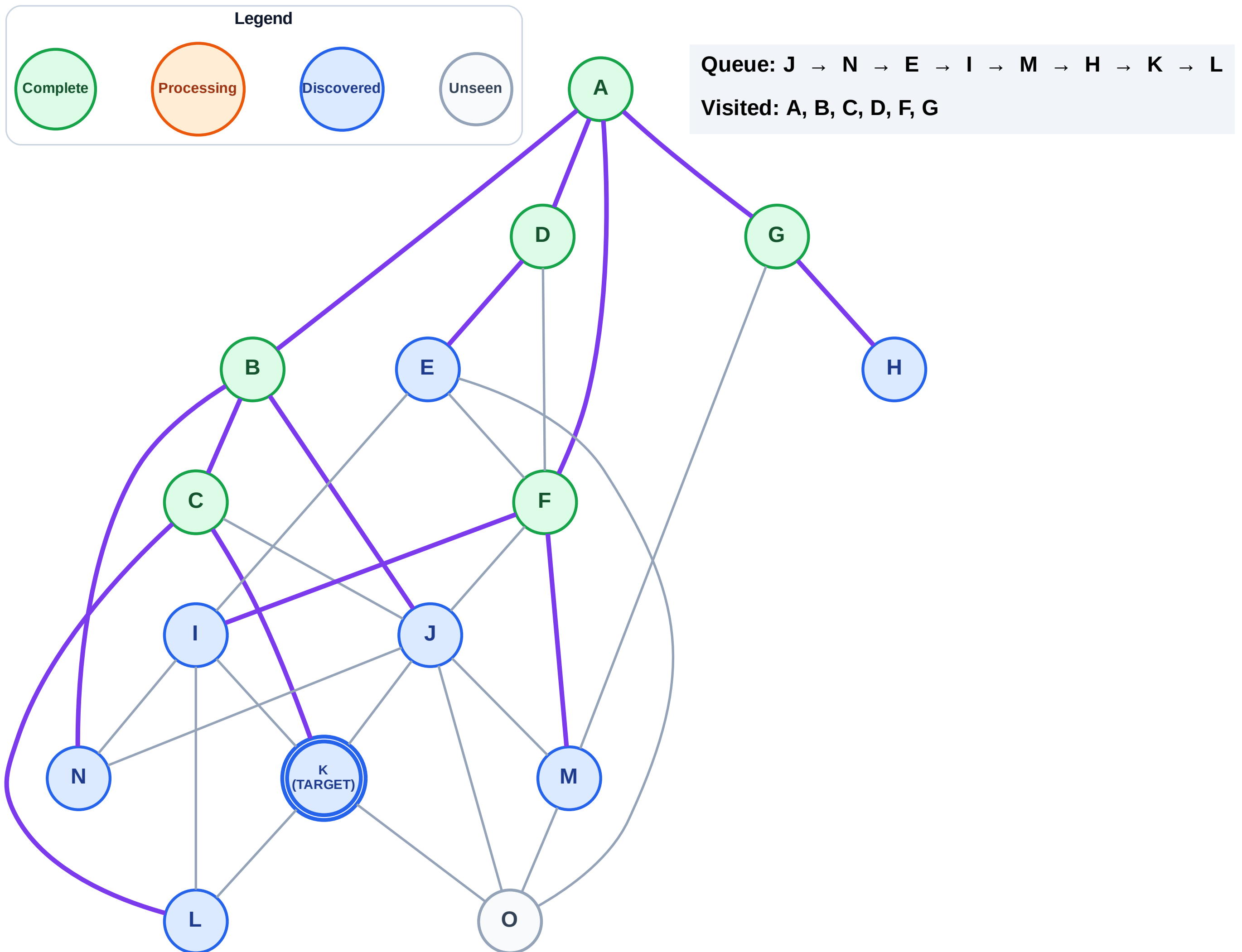
Queue: J → N → E → I → M → H

Visited: A, B, D, F, G



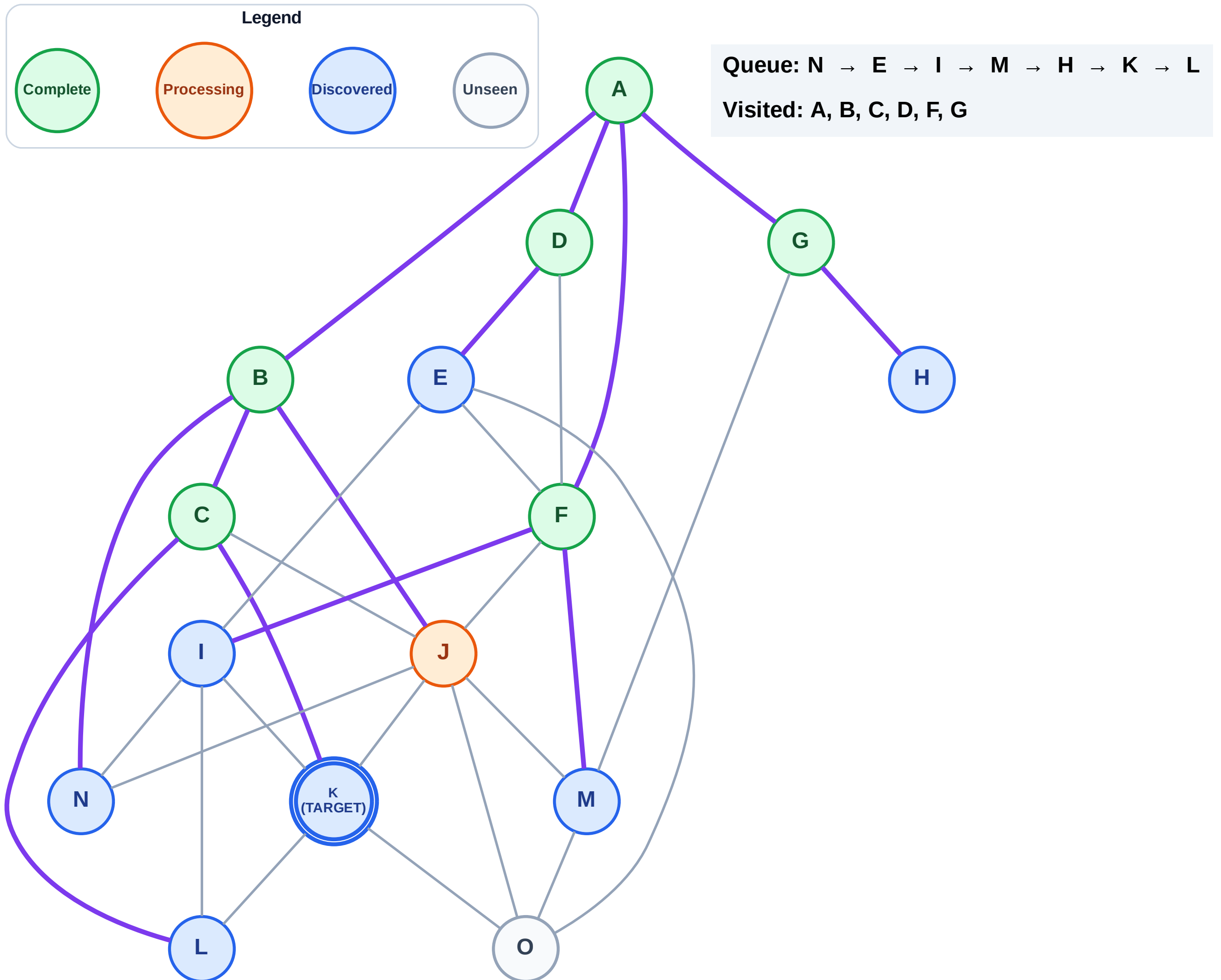
Step 13: Discover K, L from C

Step 13: Discover K, L from C



Step 14: Process node J

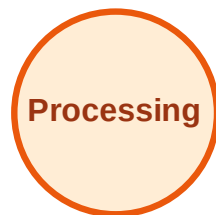
Step 14: Process node J



BFS Traversal

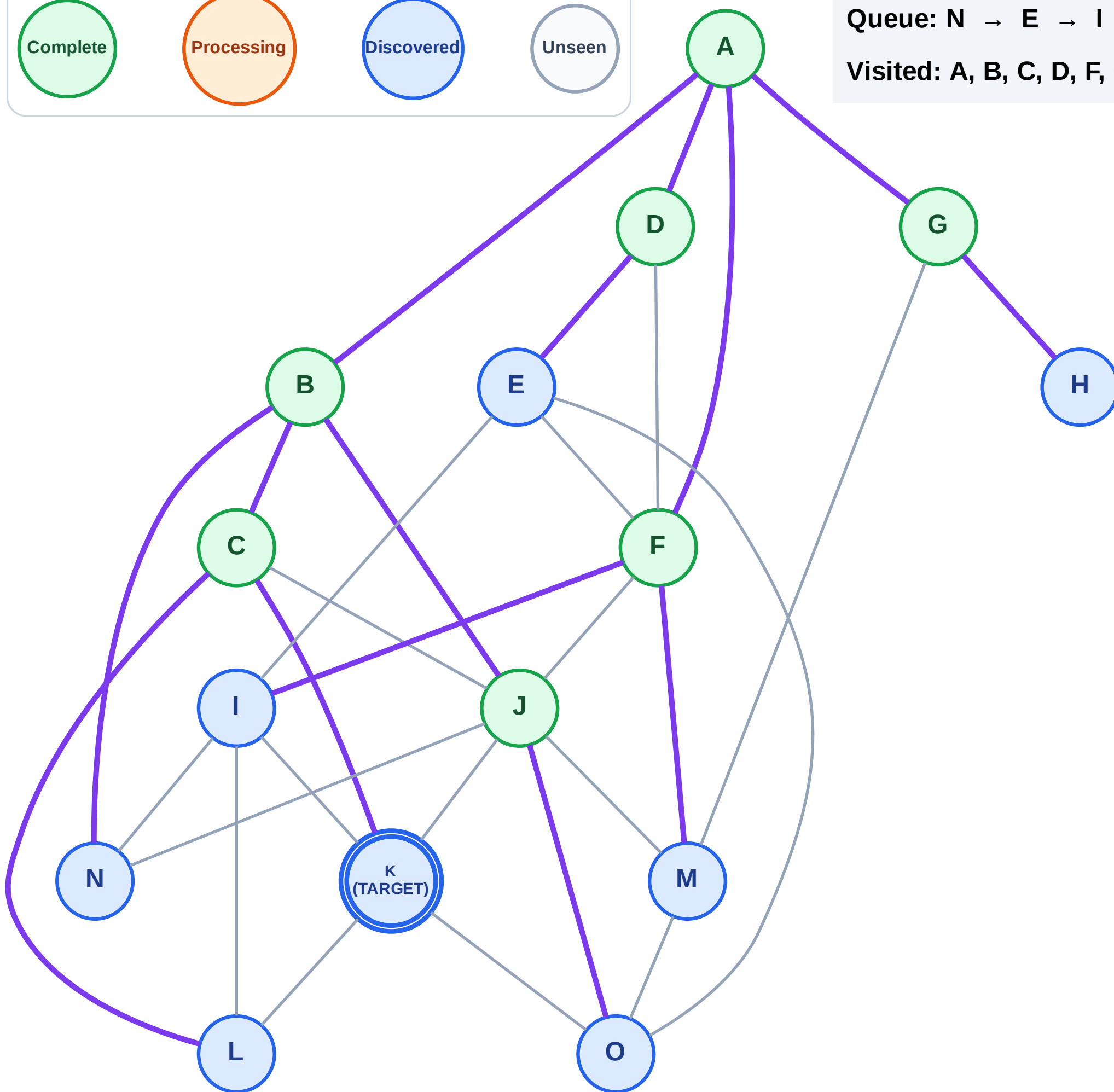
Step 15: Discover O from J

Legend



Queue: N → E → I → M → H → K → L → O

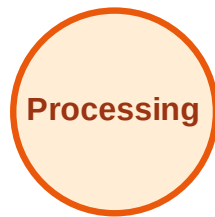
Visited: A, B, C, D, F, G, J



BFS Traversal

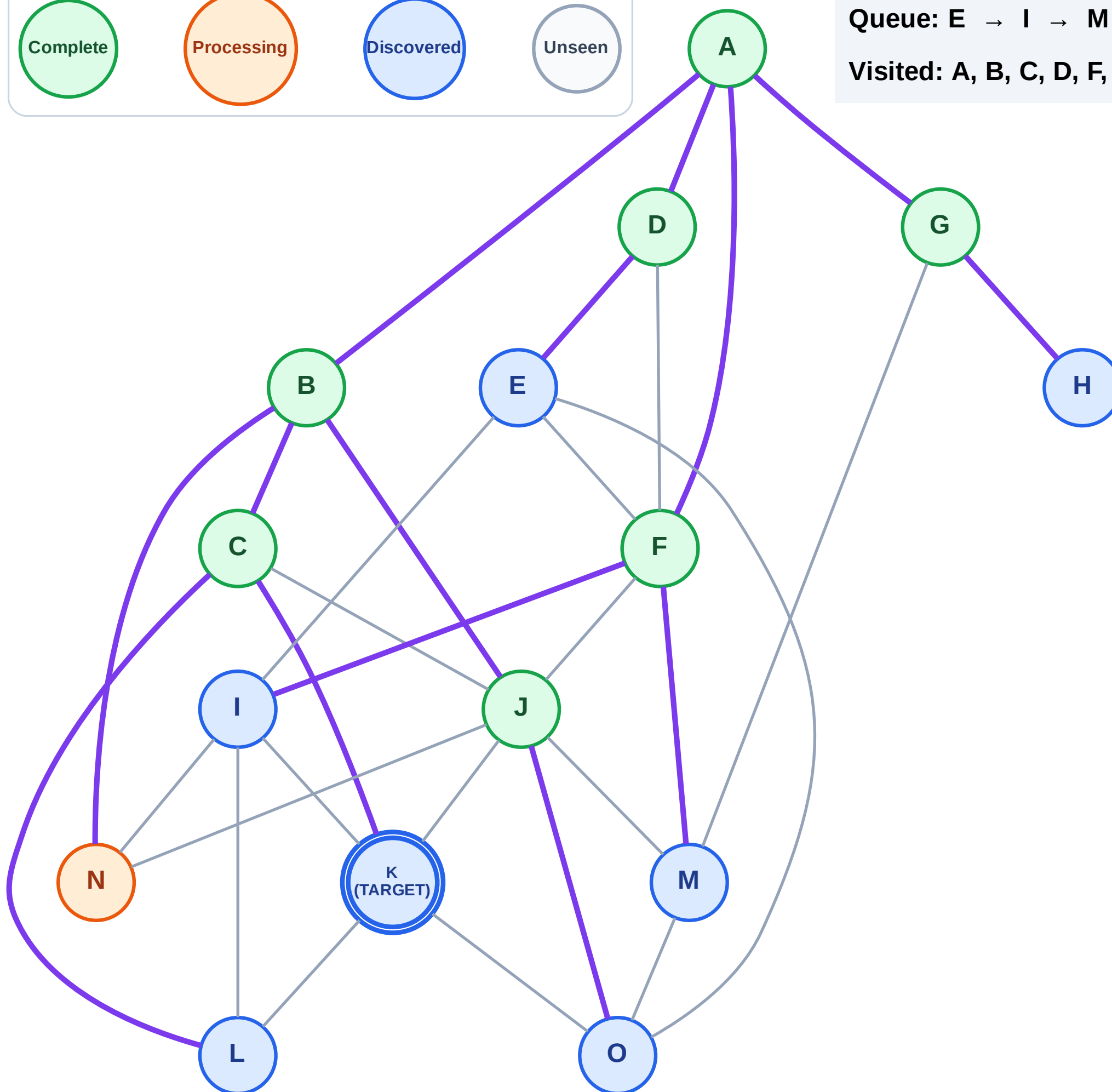
Step 16: Process node N

Legend



Queue: E → I → M → H → K → L → O

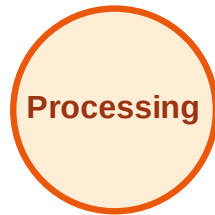
Visited: A, B, C, D, F, G, J



BFS Traversal

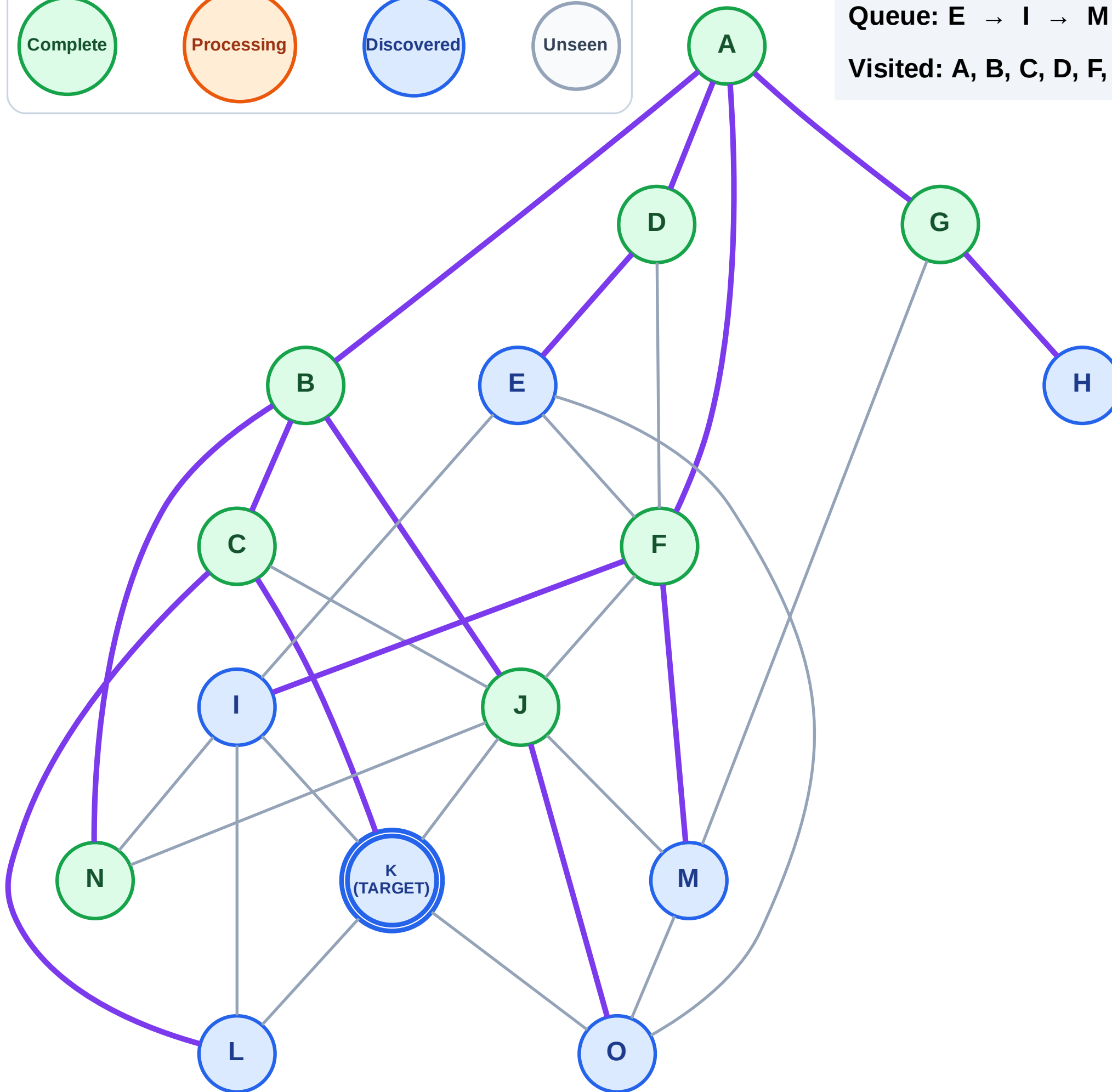
Step 17: No new neighbors from N

Legend



Queue: E → I → M → H → K → L → O

Visited: A, B, C, D, F, G, J, N



BFS Traversal

Step 18: Process node E

Legend

Complete

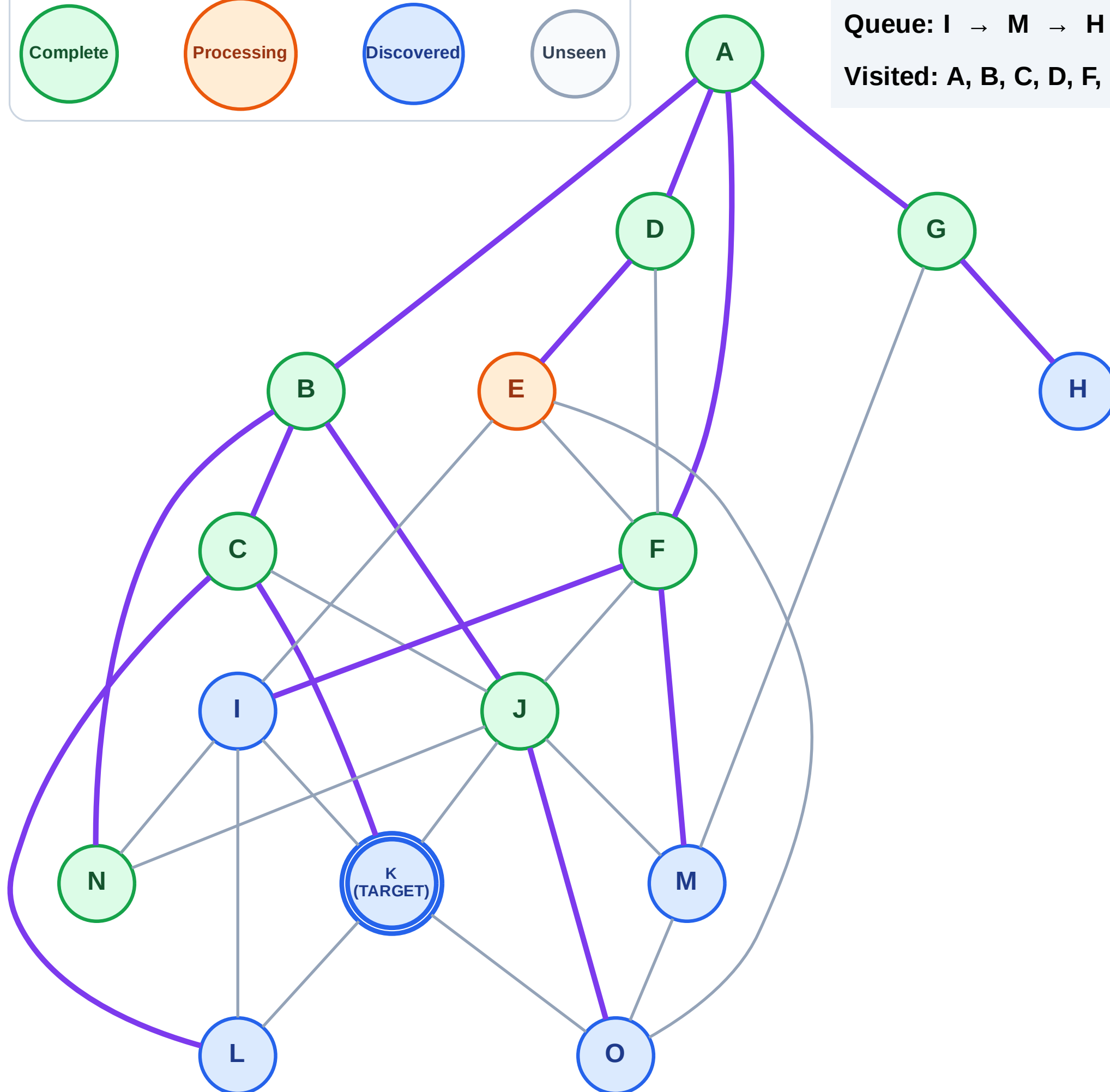
Processing

Discovered

Unseen

Queue: I → M → H → K → L → O

Visited: A, B, C, D, F, G, J, N



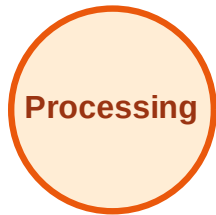
BFS Traversal

Step 19: No new neighbors from E

Legend



Complete



Processing



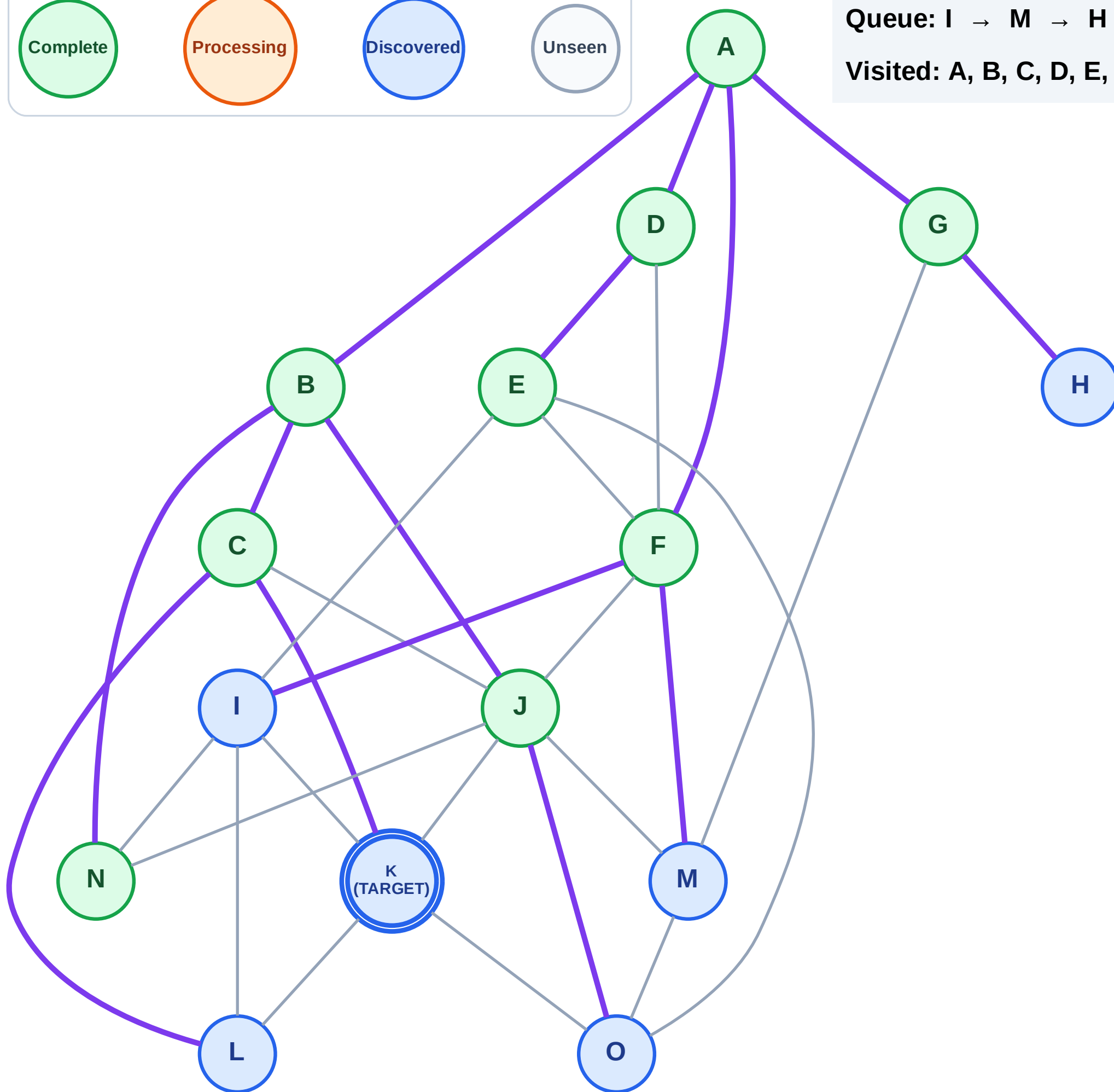
Discovered



Unseen

Queue: I → M → H → K → L → O

Visited: A, B, C, D, E, F, G, J, N



BFS Traversal

Step 20: Process node I

Legend

Complete

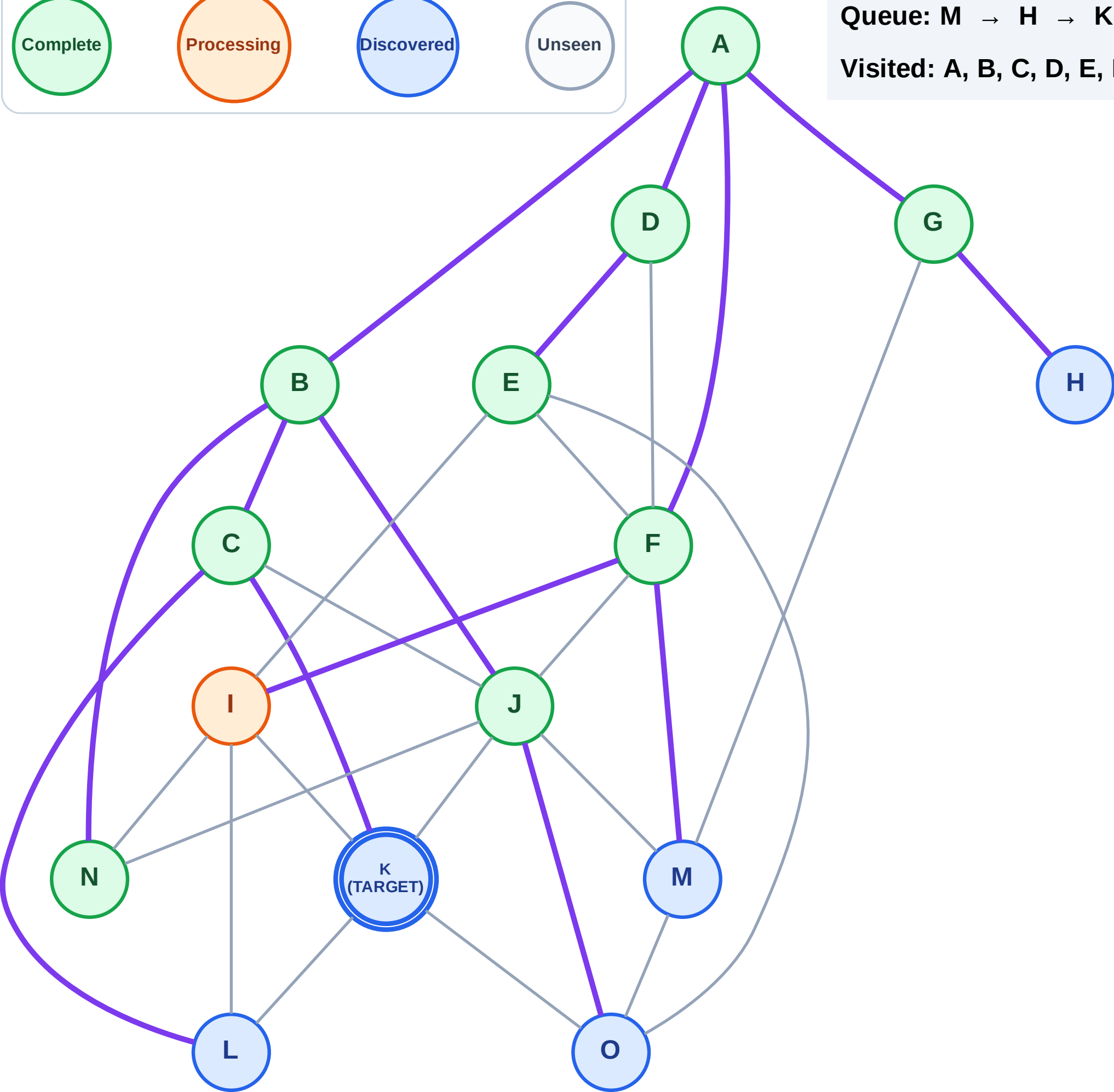
Processing

Discovered

Unseen

Queue: M → H → K → L → O

Visited: A, B, C, D, E, F, G, J, N



BFS Traversal

Step 21: No new neighbors from I

Legend



Complete



Processing



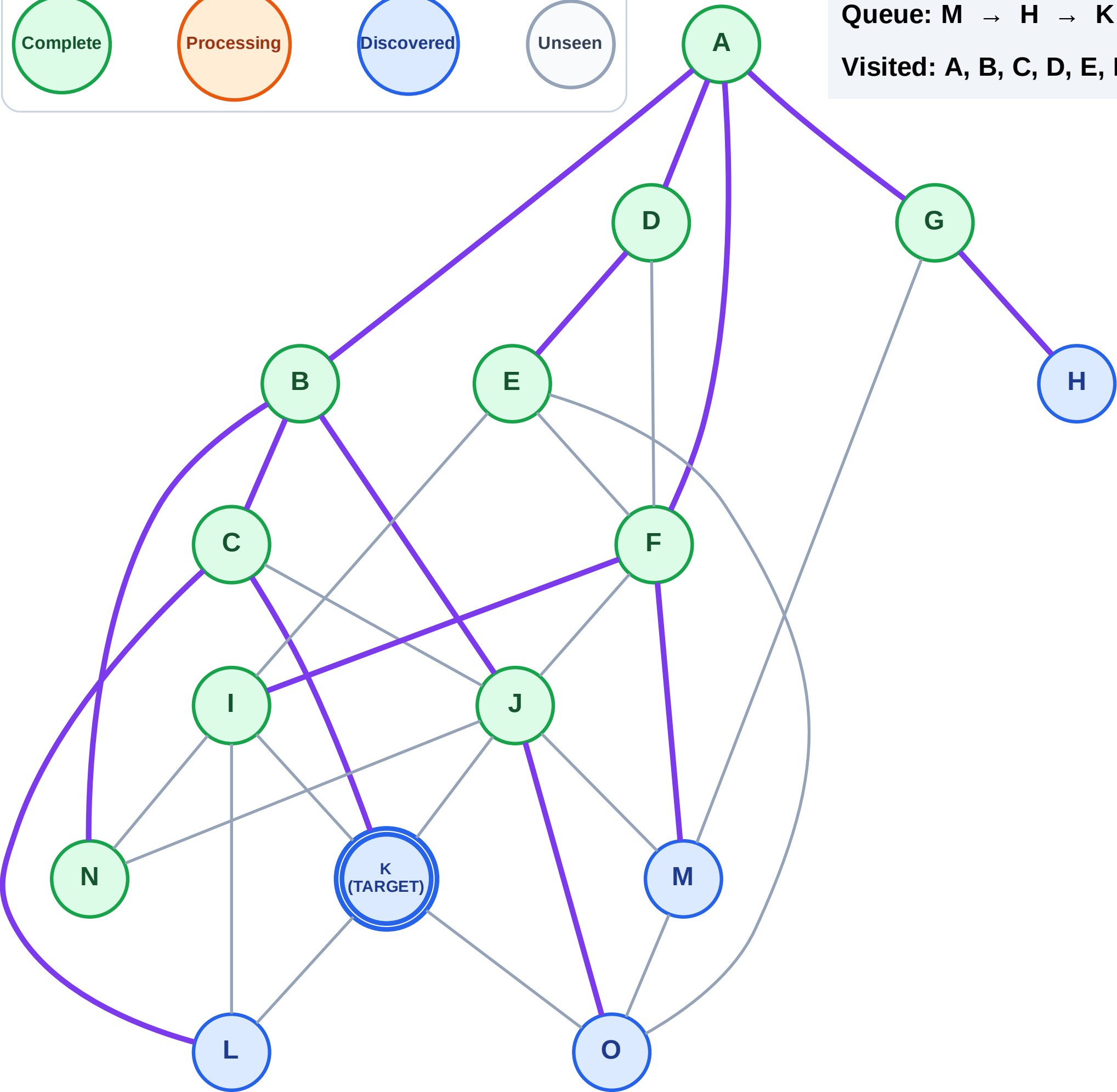
Discovered



Unseen

Queue: M → H → K → L → O

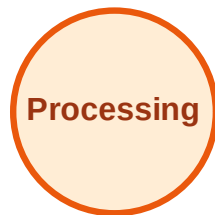
Visited: A, B, C, D, E, F, G, I, J, N



BFS Traversal

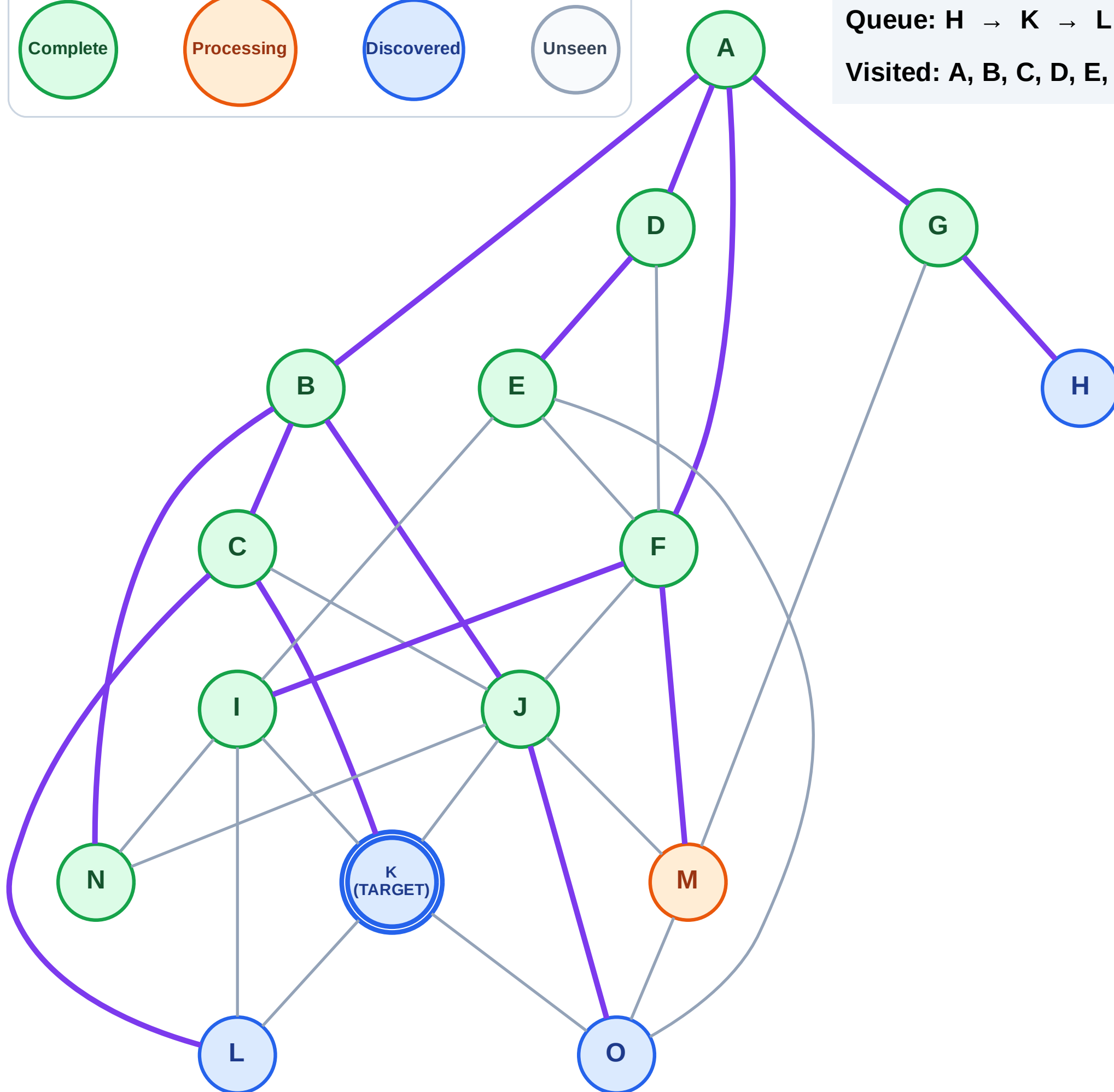
Step 22: Process node M

Legend



Queue: H → K → L → O

Visited: A, B, C, D, E, F, G, I, J, N



BFS Traversal

Step 23: No new neighbors from M

Legend

Complete

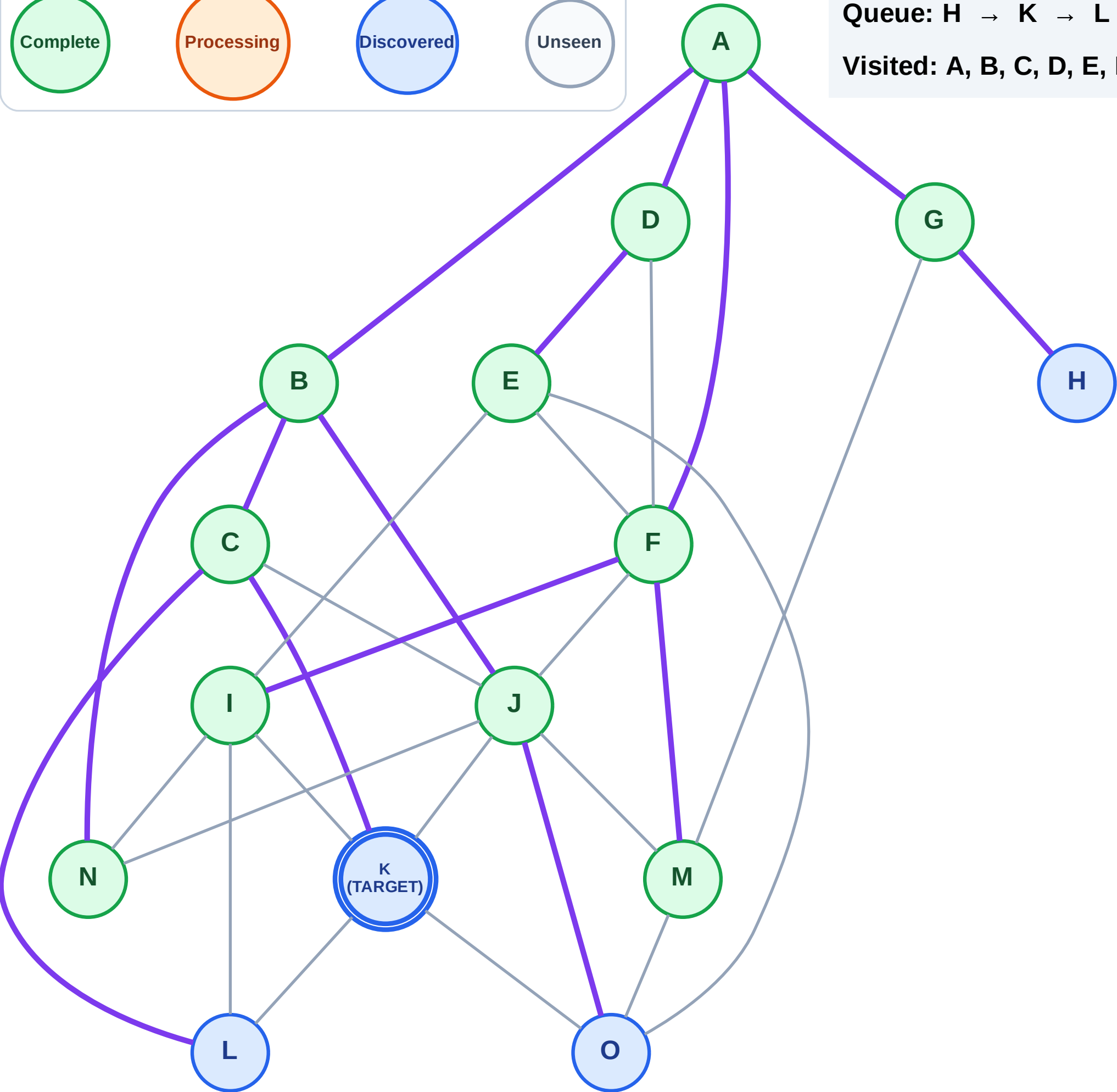
Processing

Discovered

Unseen

Queue: H → K → L → O

Visited: A, B, C, D, E, F, G, I, J, M, N



BFS Traversal

Step 24: Process node H

Legend

Complete

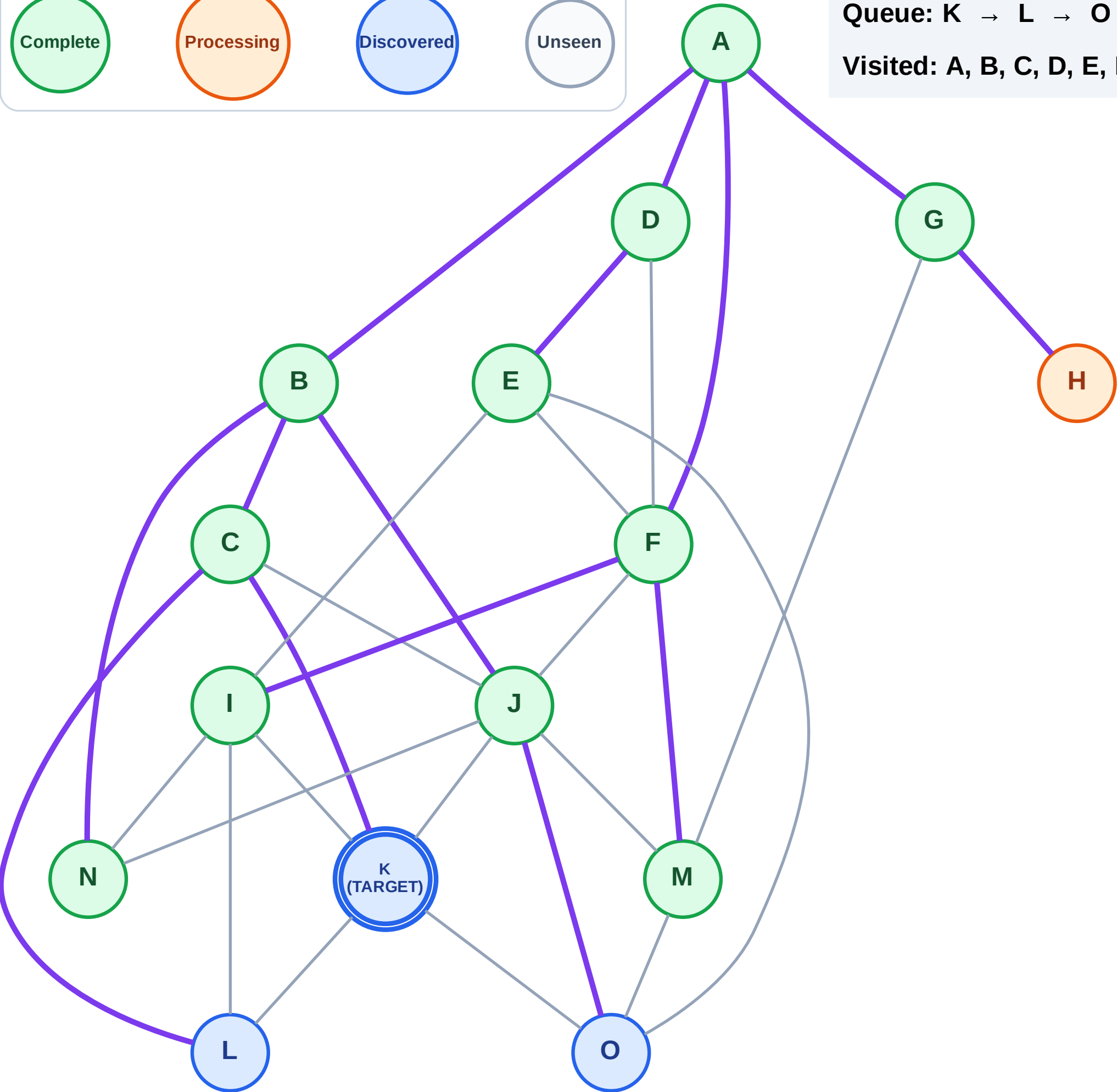
Processing

Discovered

Unseen

Queue: K → L → O

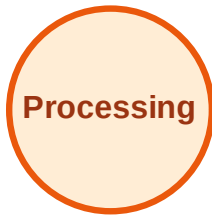
Visited: A, B, C, D, E, F, G, I, J, M, N



BFS Traversal

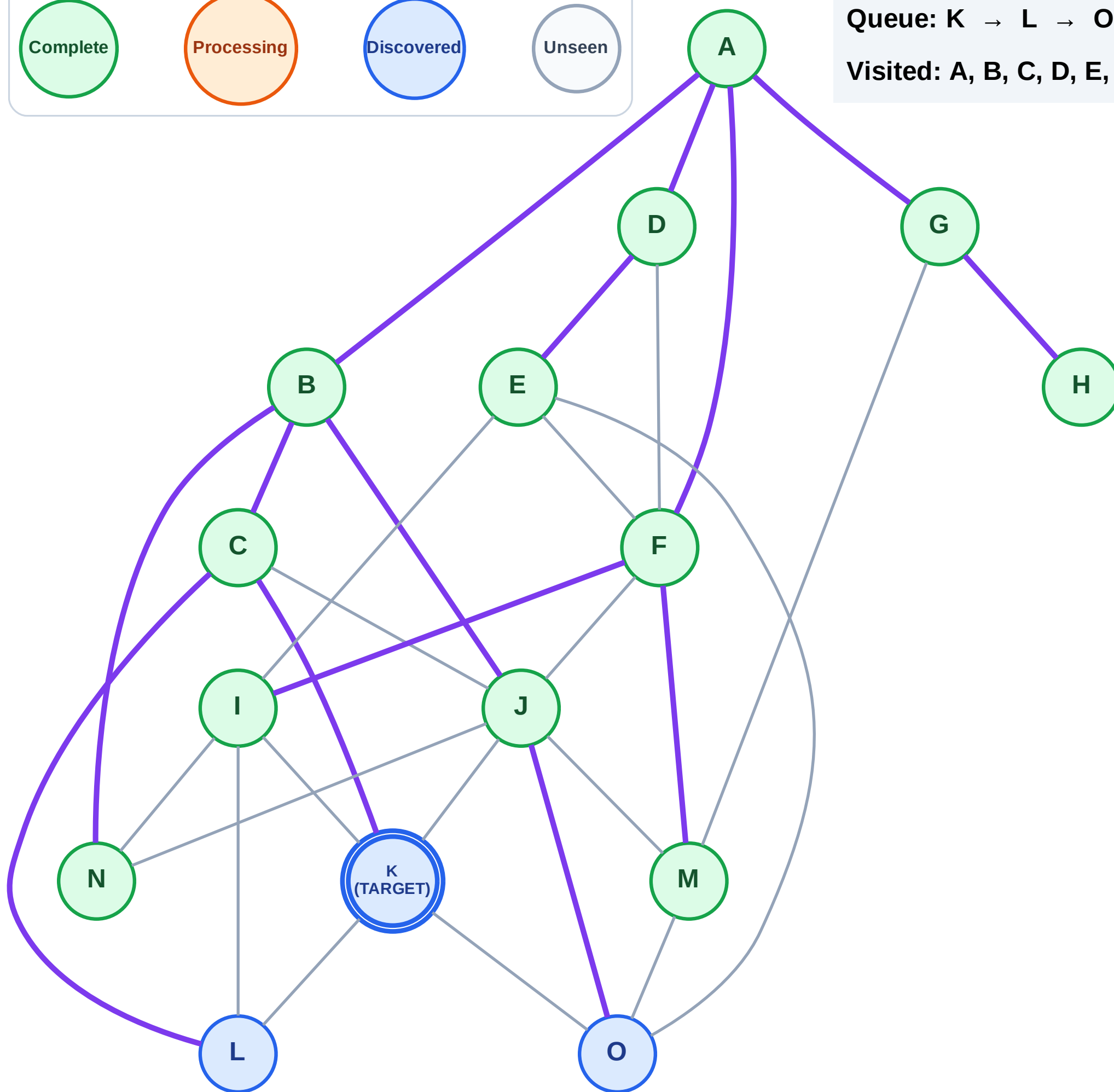
Step 25: No new neighbors from H

Legend



Queue: K → L → O

Visited: A, B, C, D, E, F, G, H, I, J, M, N



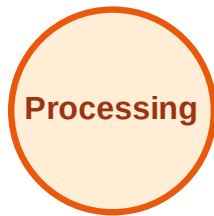
BFS Traversal

Step 26: Process node K

Legend



Complete



Processing



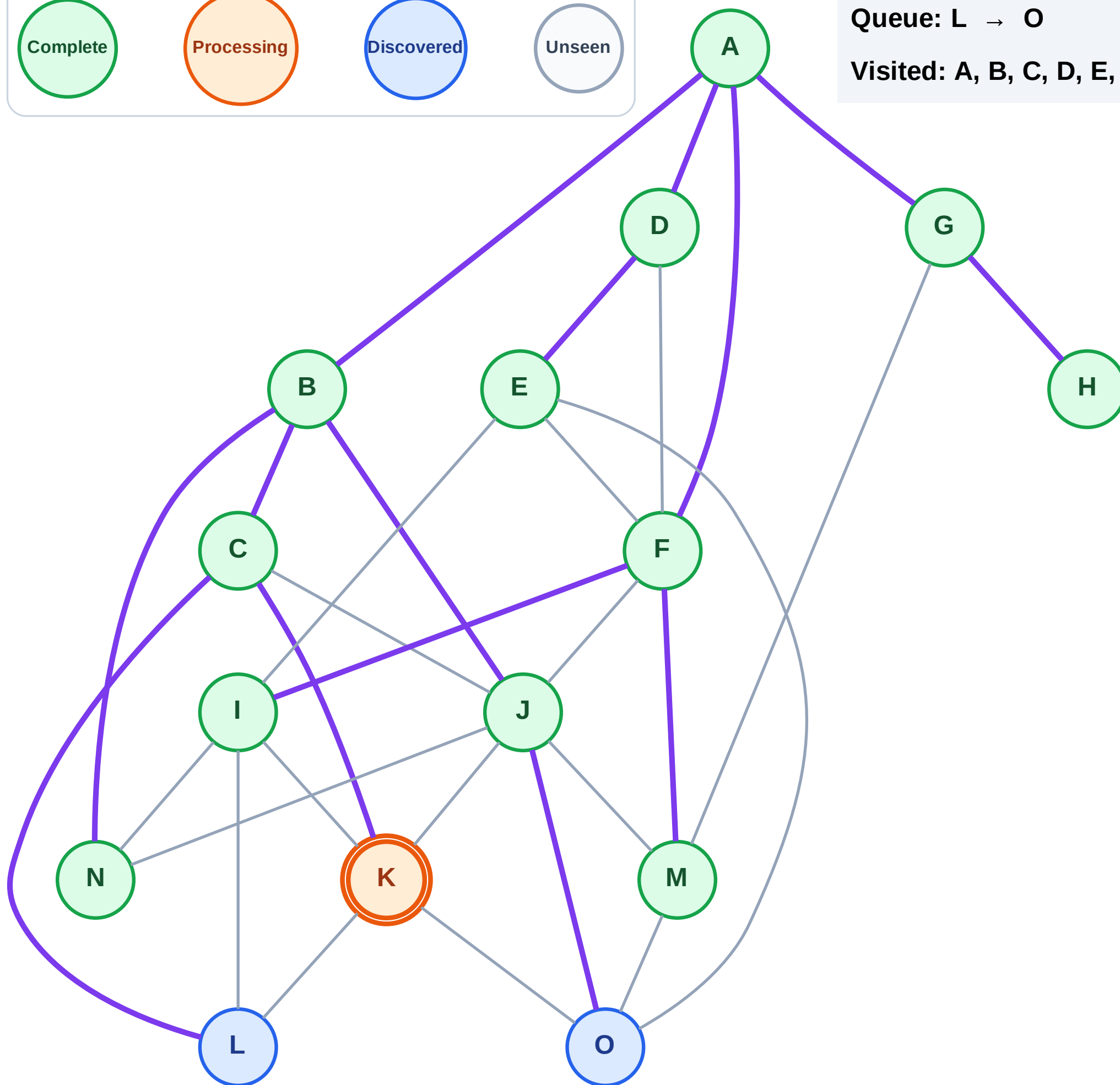
Discovered



Unseen

Queue: L → O

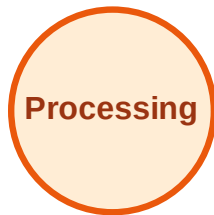
Visited: A, B, C, D, E, F, G, H, I, J, M, N



BFS Traversal

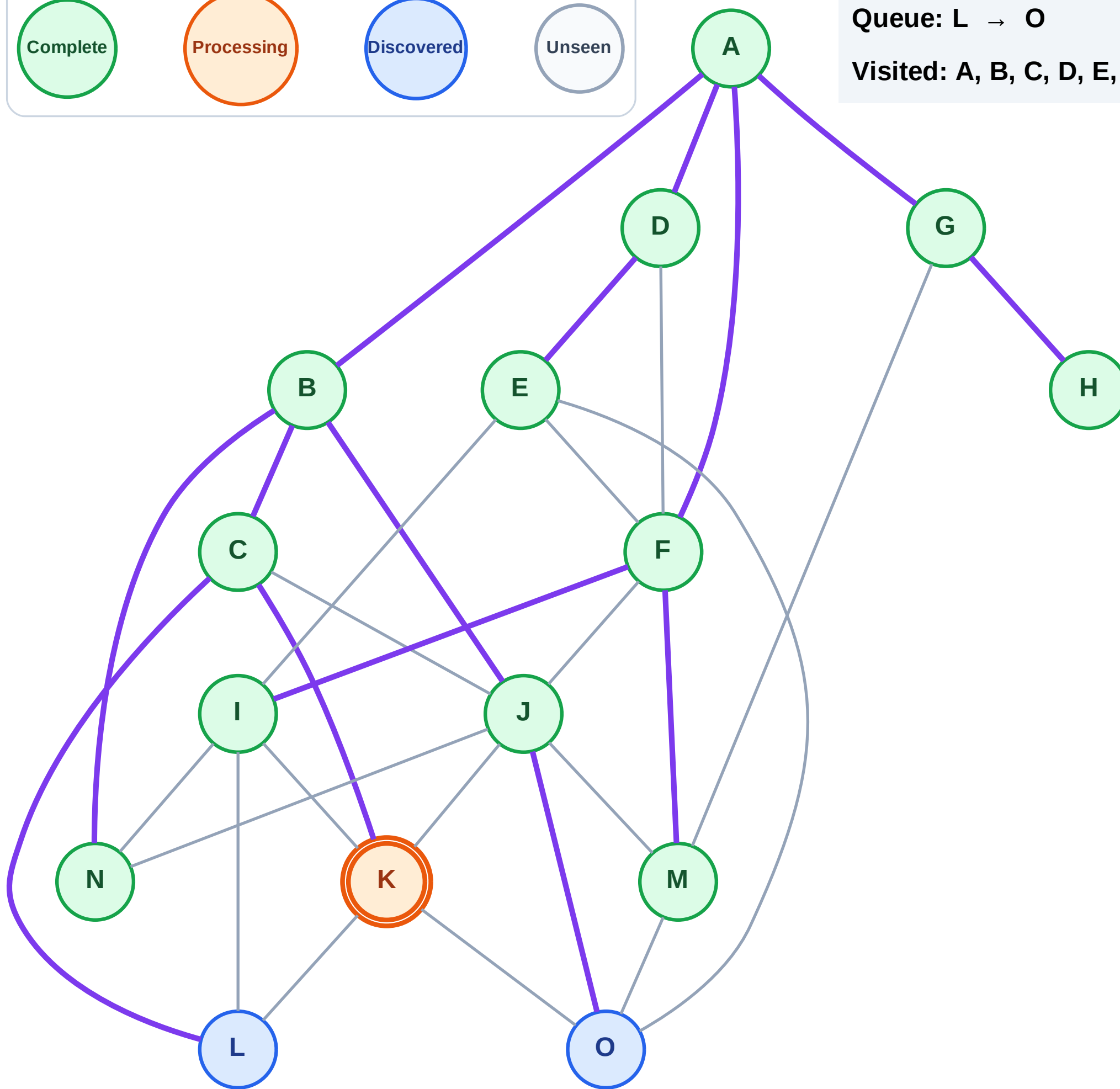
Step 27: TARGET NODE K REACHED! Path (3 hops): A → B → C → K

Legend



Queue: L → O

Visited: A, B, C, D, E, F, G, H, I, J, K, M, N



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

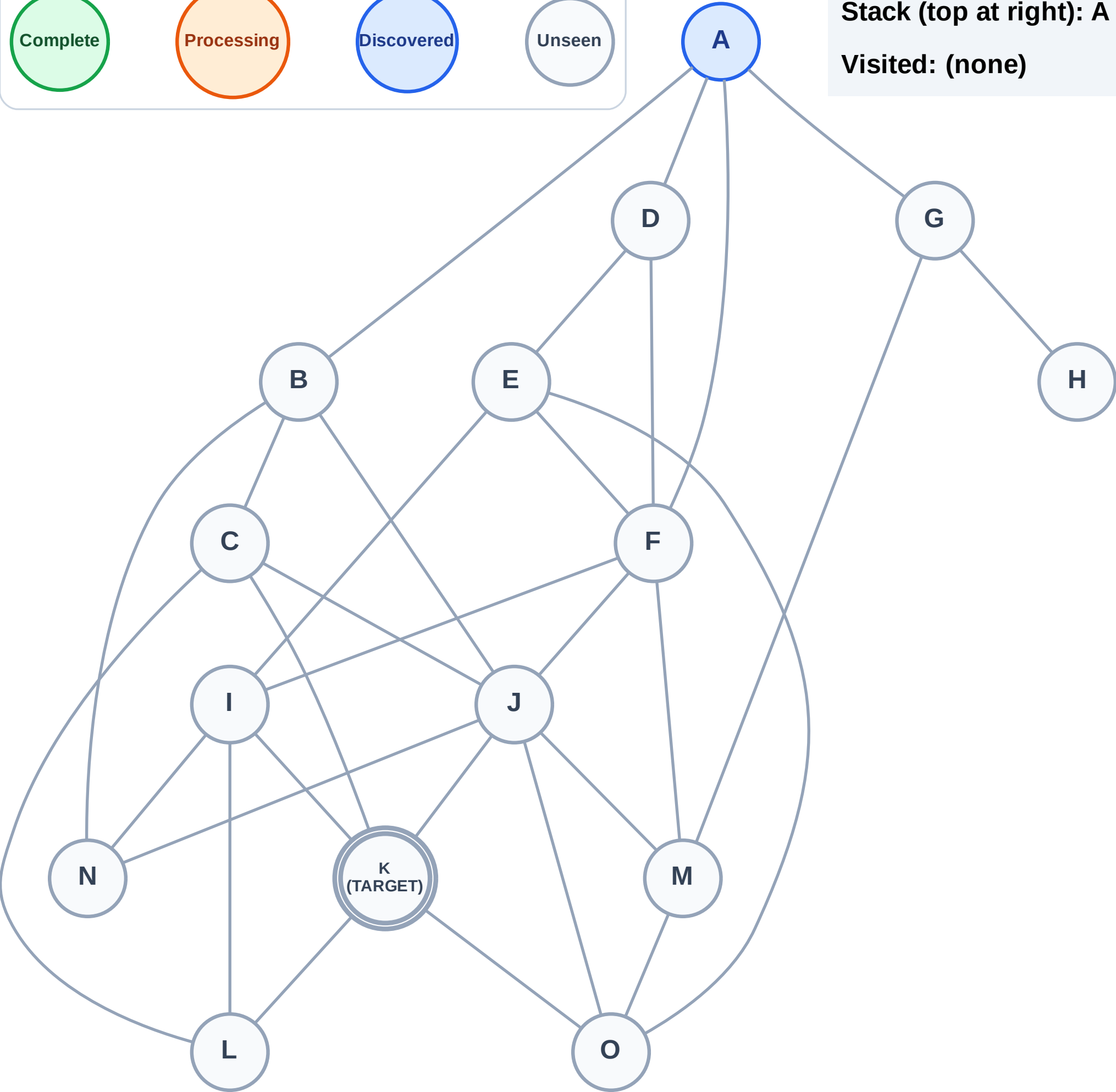
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

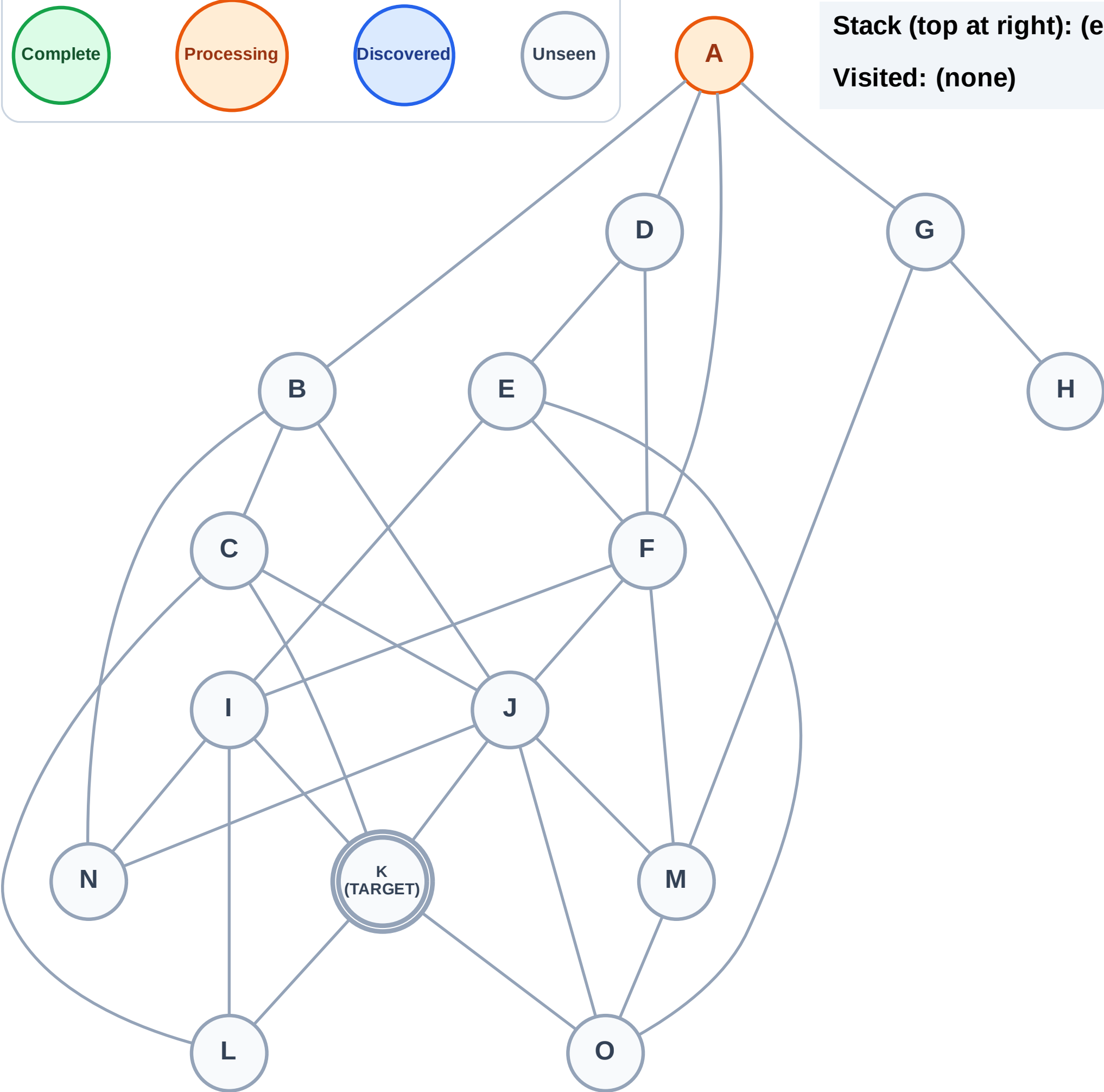
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover G, F, D, B from A

Legend

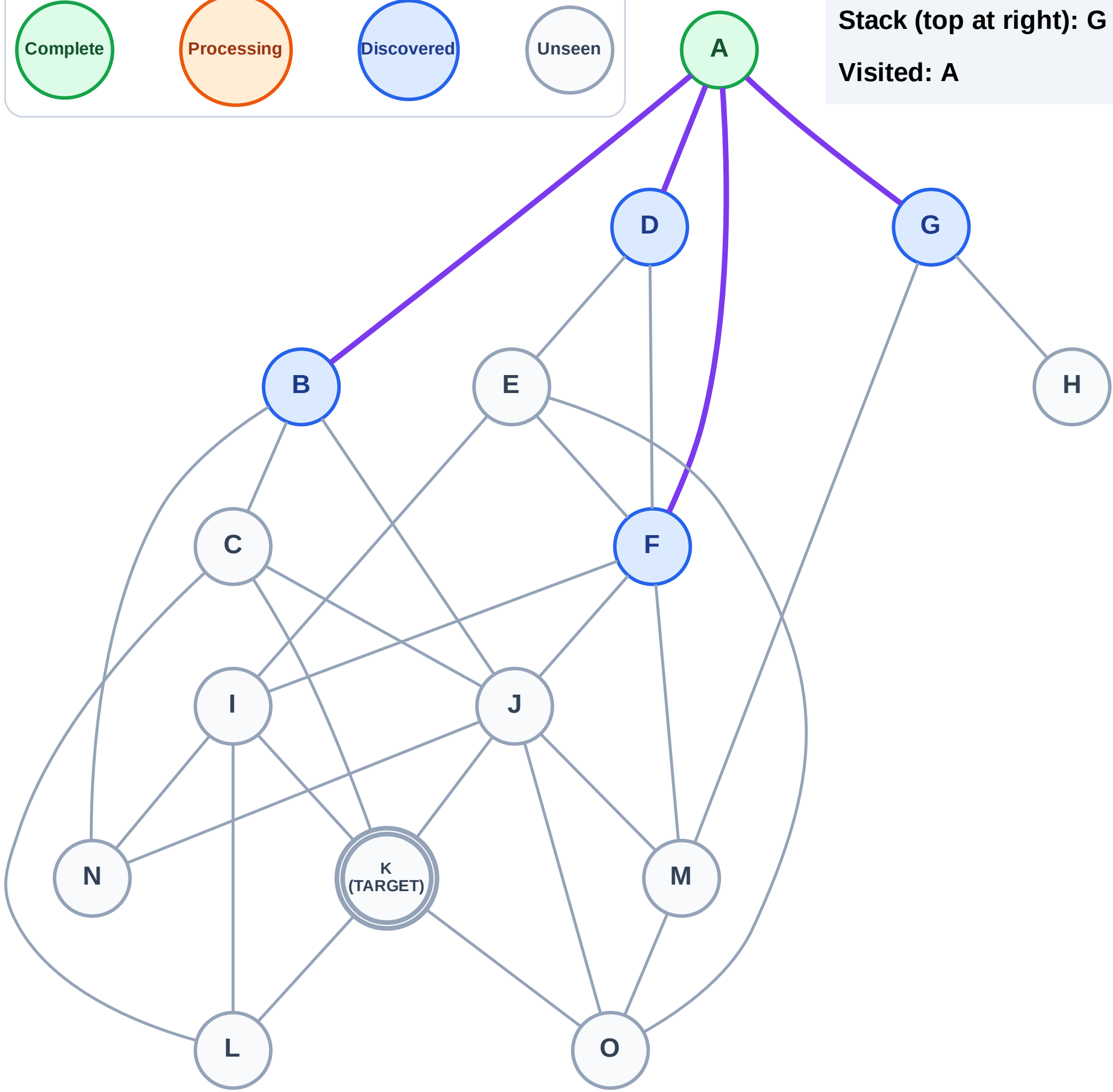
Complete

Processing

Discovered

Unseen

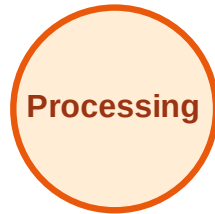
Stack (top at right): G | F | D | B
Visited: A



DFS Traversal

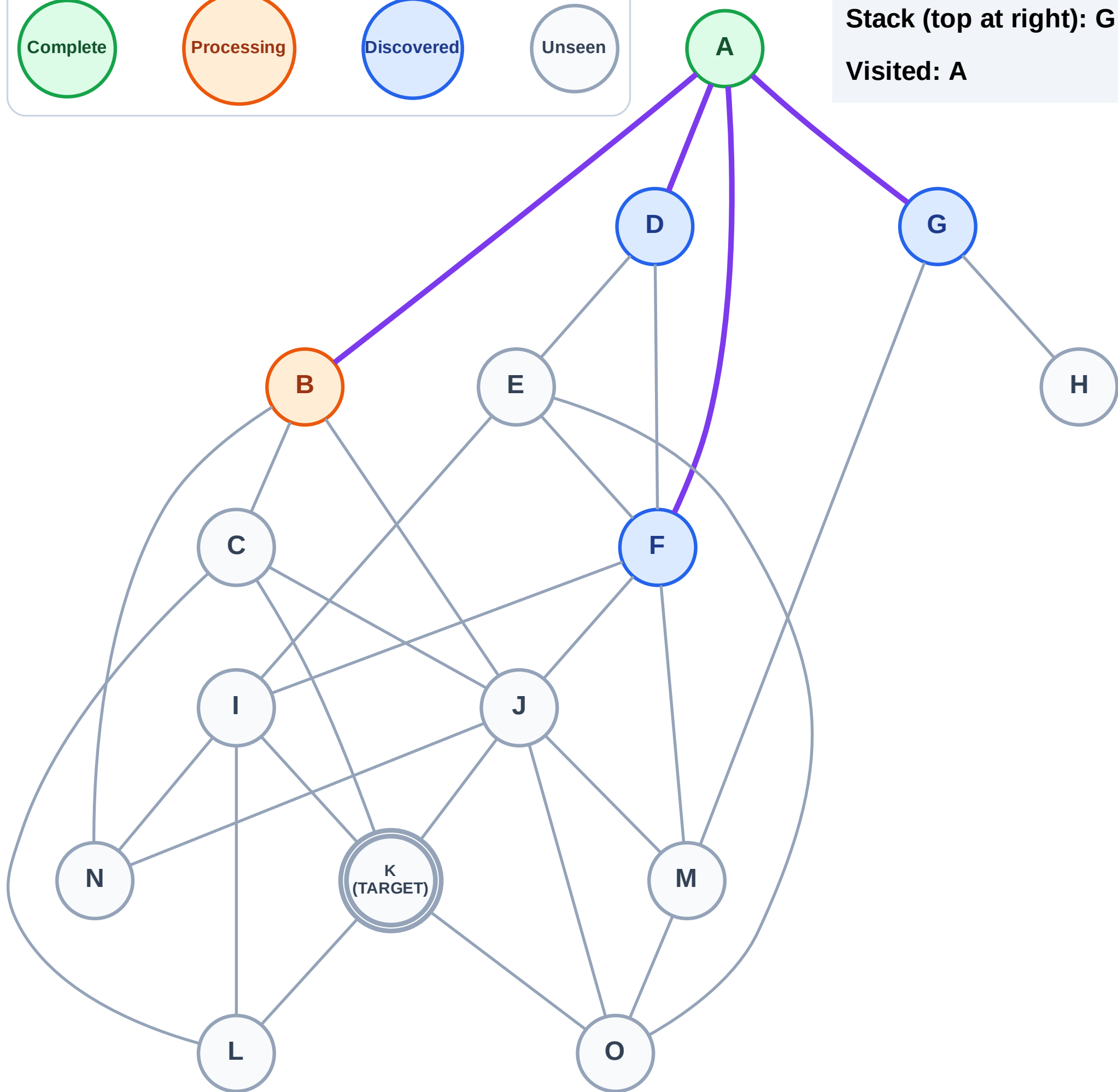
Step 4: Process node B

Legend



Stack (top at right): G | F | D

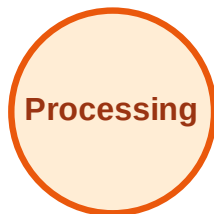
Visited: A



DFS Traversal

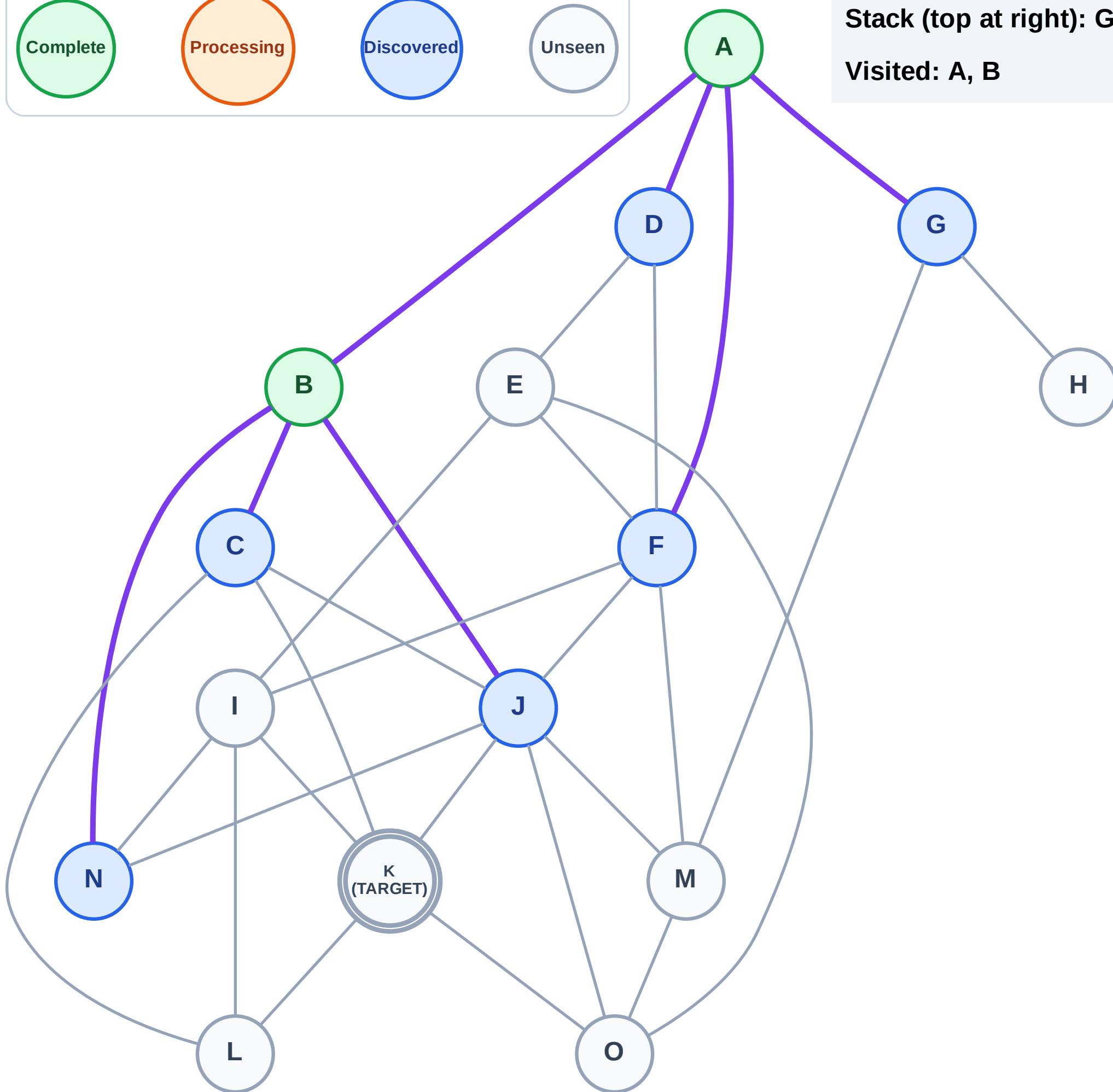
Step 5: Discover N, J, C from B

Legend



Stack (top at right): G | F | D | N | J | C

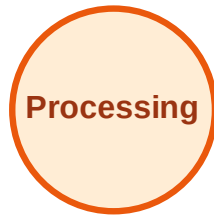
Visited: A, B



DFS Traversal

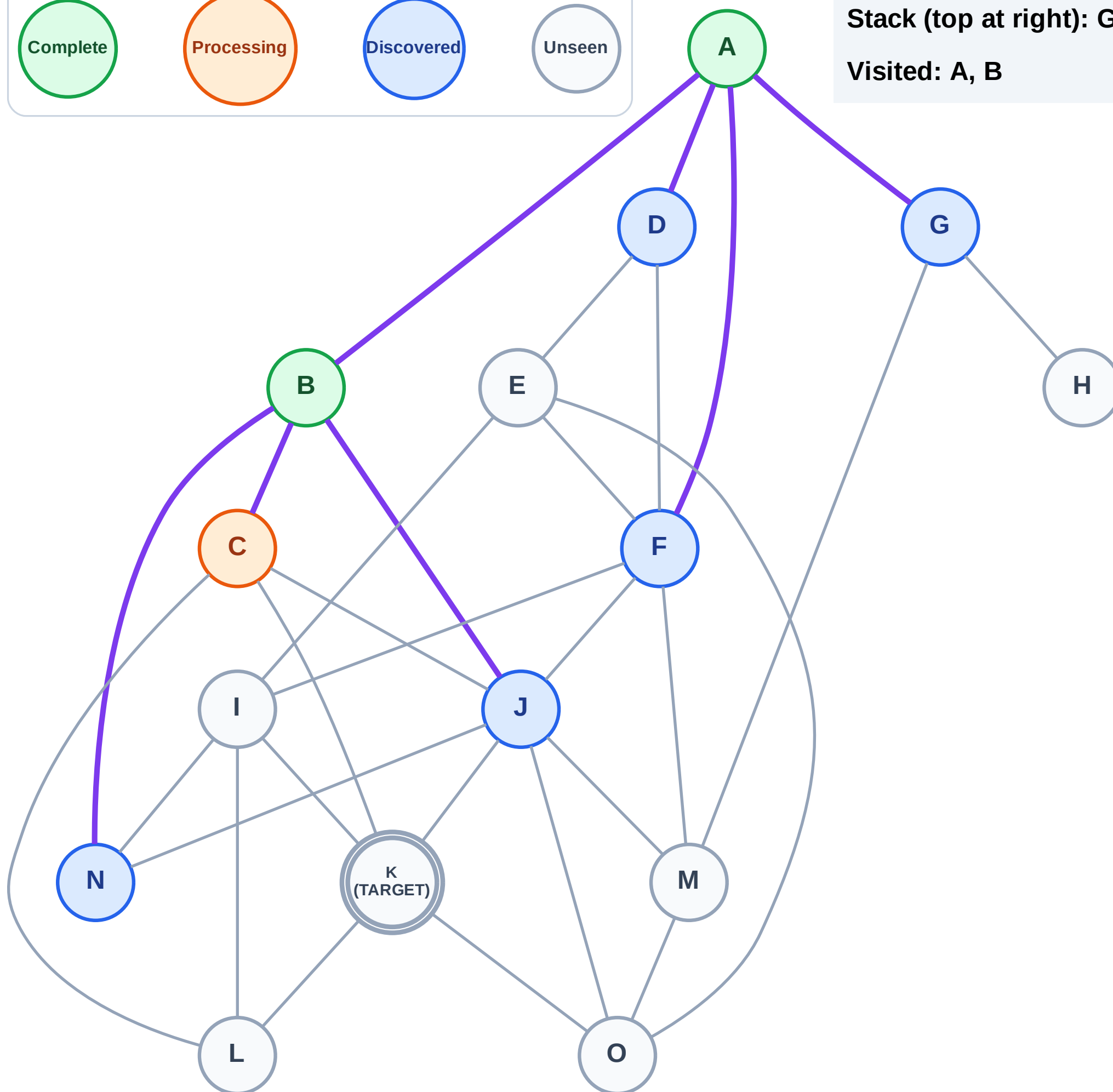
Step 6: Process node C

Legend



Stack (top at right): G | F | D | N | J

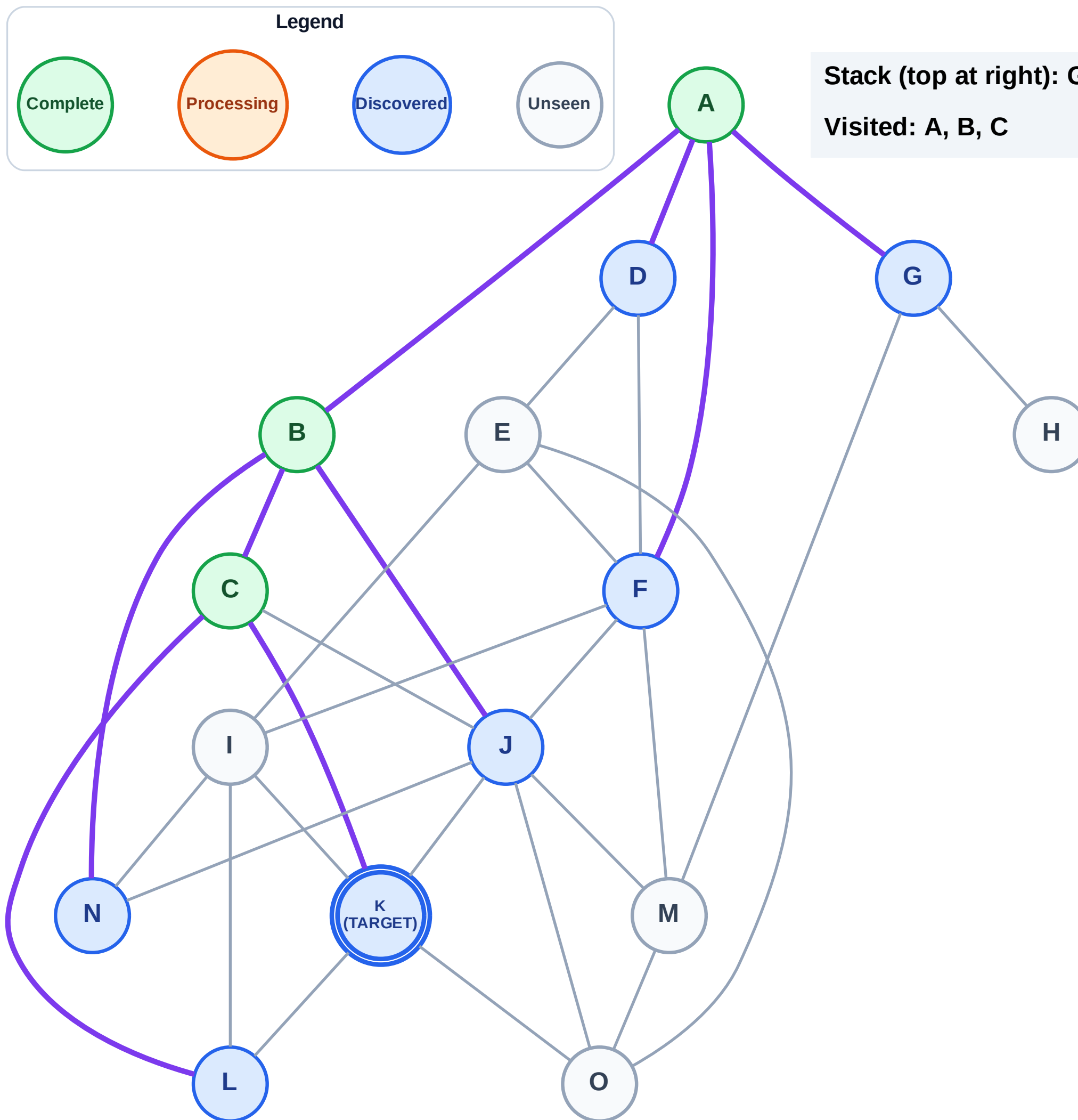
Visited: A, B



Step 7: Discover L, K from C

Stack (top at right): G | F | D | N | J | L | K

Visited: A, B, C



DFS Traversal

Step 8: Process node K

Legend

Complete

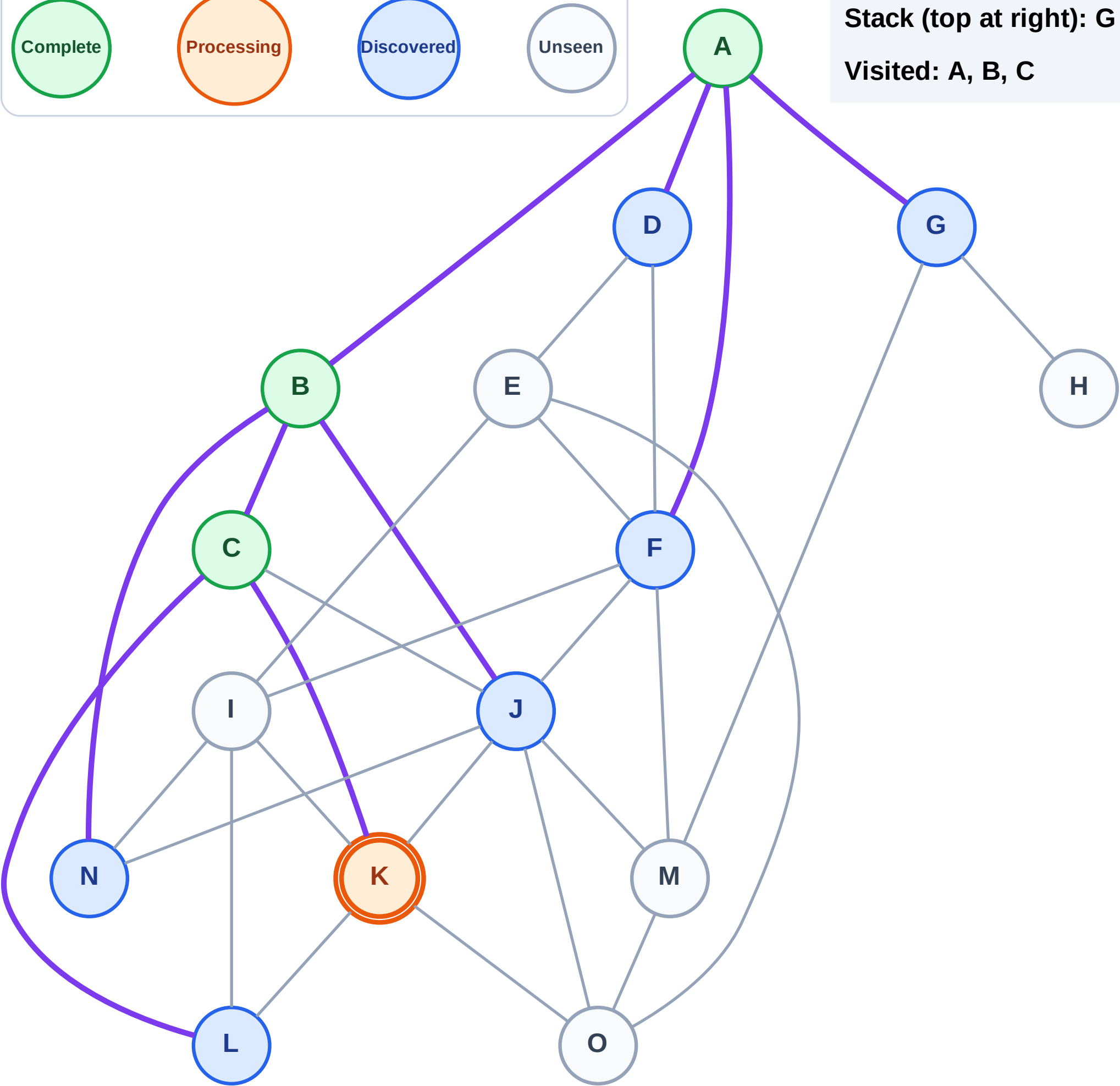
Processing

Discovered

Unseen

Stack (top at right): G | F | D | N | J | L

Visited: A, B, C



DFS Traversal

Step 9: TARGET NODE K REACHED! Path (3 hops): A → B → C → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): G | F | D | N | J | L
Visited: A, B, C, K

